

Lecture 2

Thermodynamics from lattice dynamics



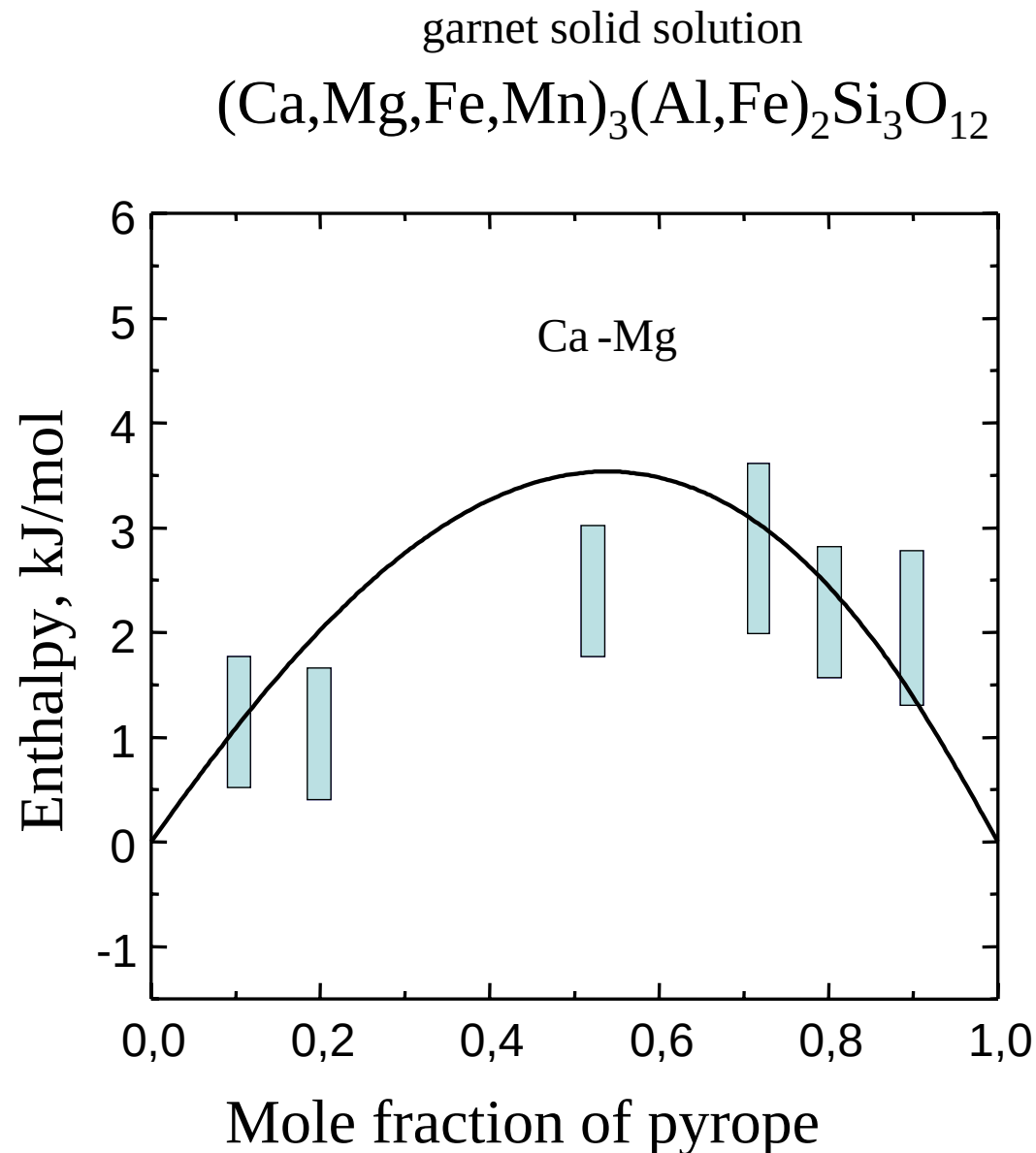
Outline of Lecture 2

- Why do we need computer simulations?
- Empirical pair potentials and SLEC
- The effect of pressure
- The effect of temperature
- P-T phase diagrams



The need for thermodynamic data

Metamorphic petrology



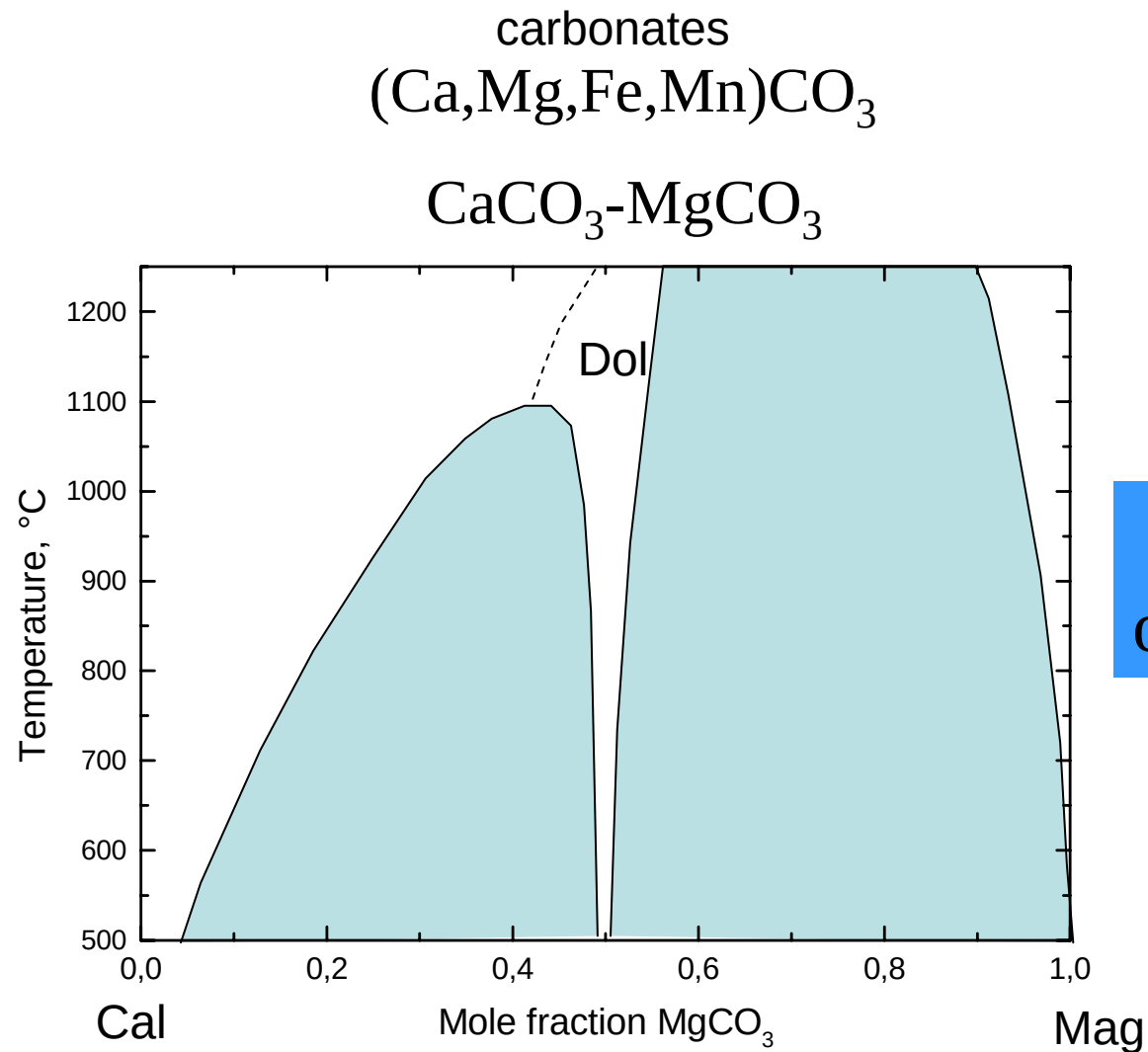
Experimental data:
Newton et al (1977)
for the Ca-Mg series

No good data for:

Ca - Fe
Ca - Mn
Mg - Mn
Fe - Mg
Fe - Mn

The need for thermodynamic data

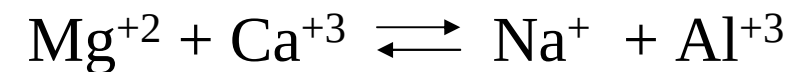
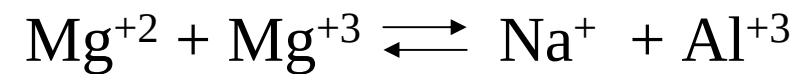
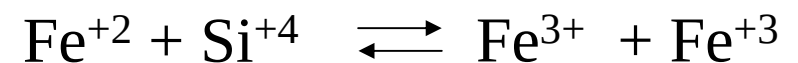
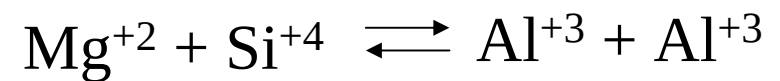
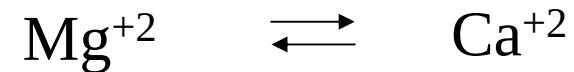
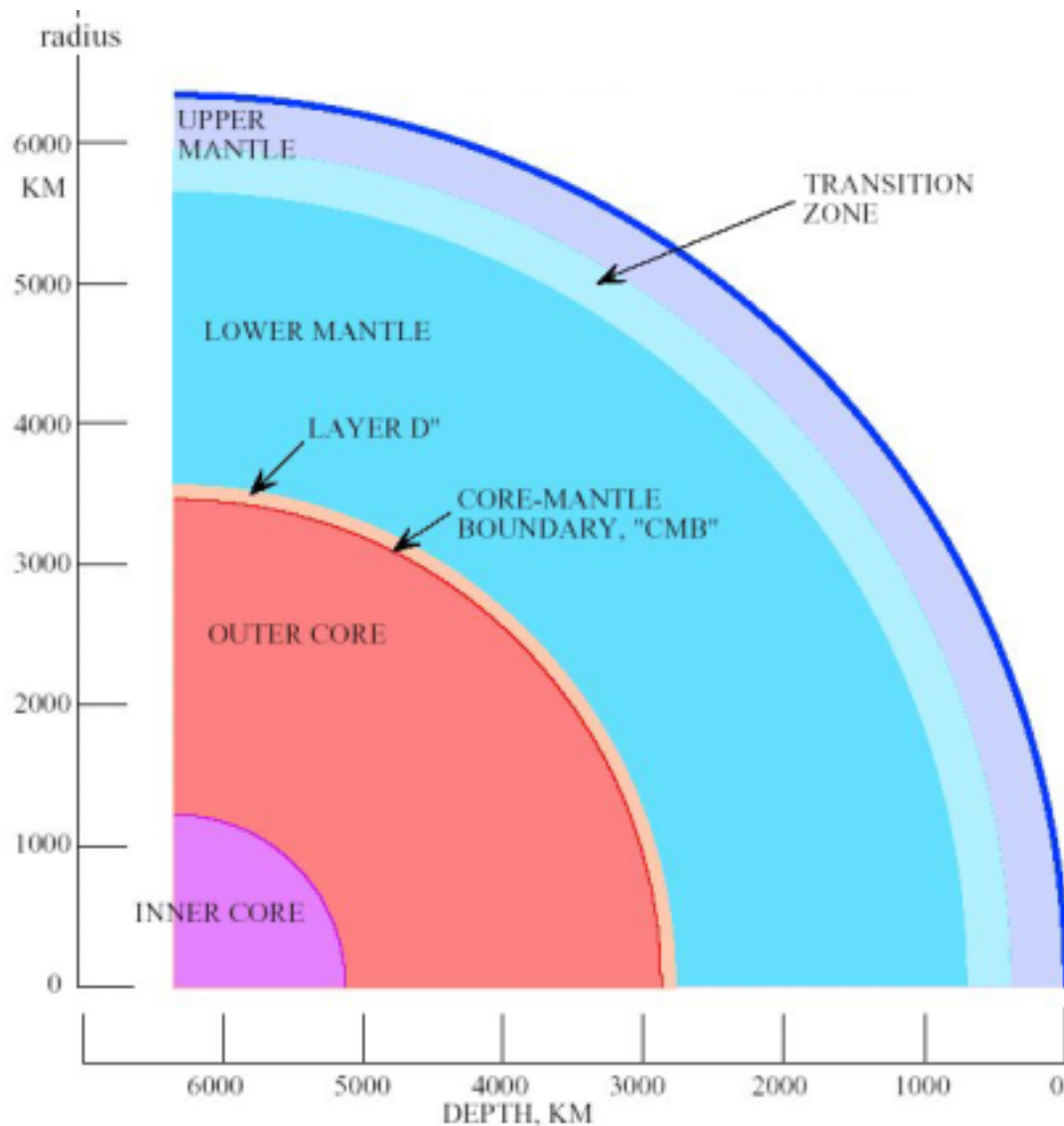
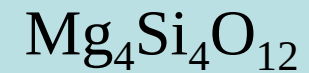
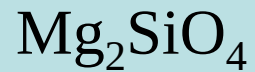
Metamorphic petrology



No good data for
other binary systems

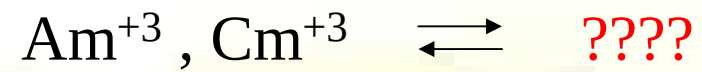
Goldsmith & Heard, 1961

Mantle mineralogy



No good data for
solid solutions

Radioactive waste



No good data for
solid solutions

Thermodynamics of melts



Almost no data

Simulation methods

Empirical
interatomic
potentials

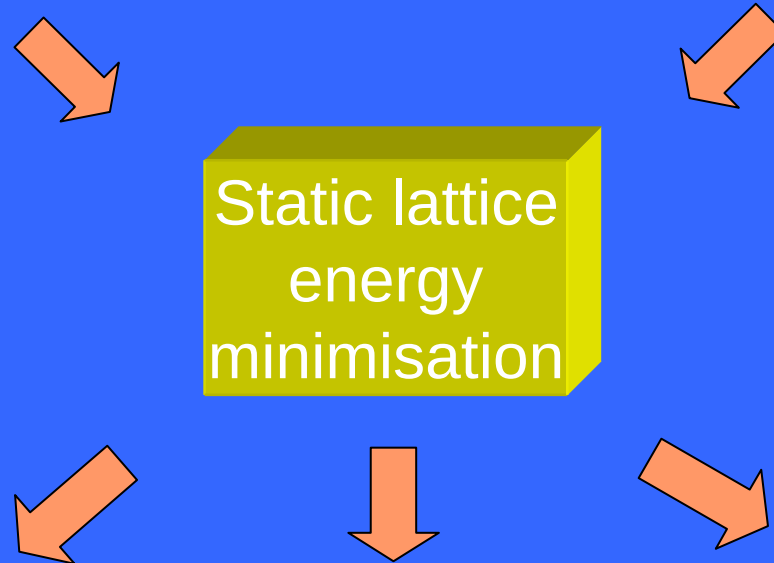
Quantum
mechanics

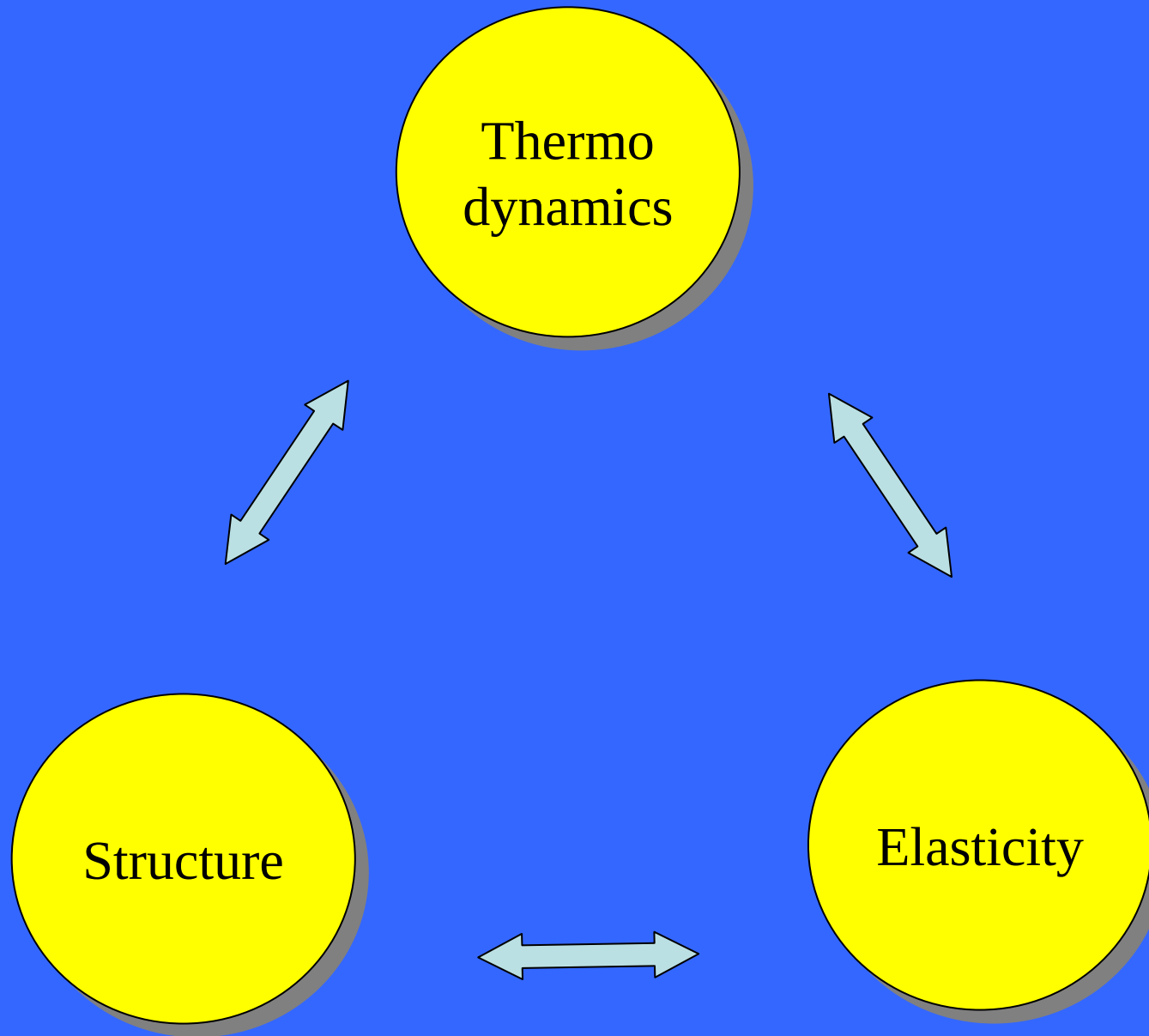
Static lattice
energy
minimisation

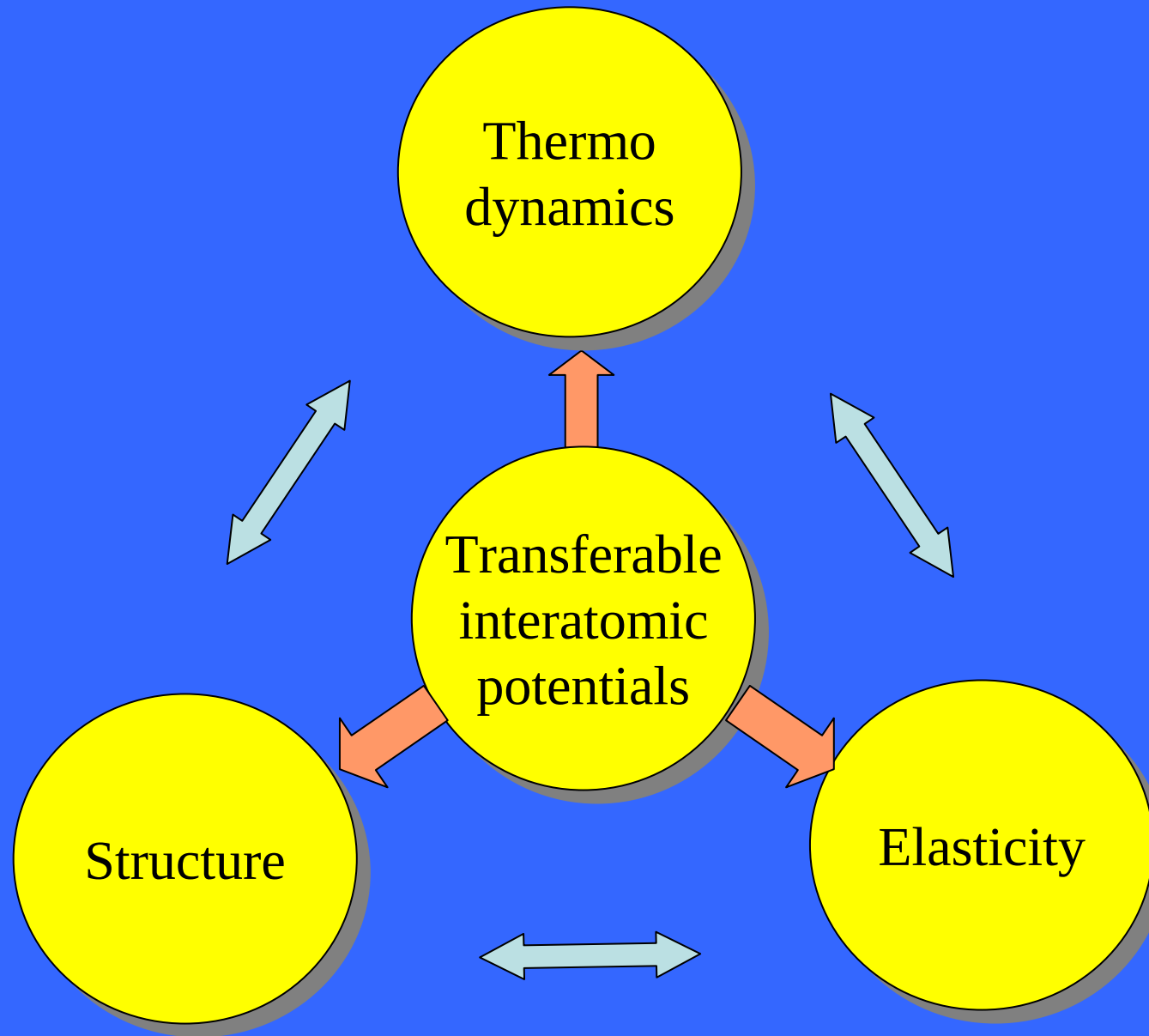
Lattice
dynamics

Monte Carlo

Molecular
dynamics





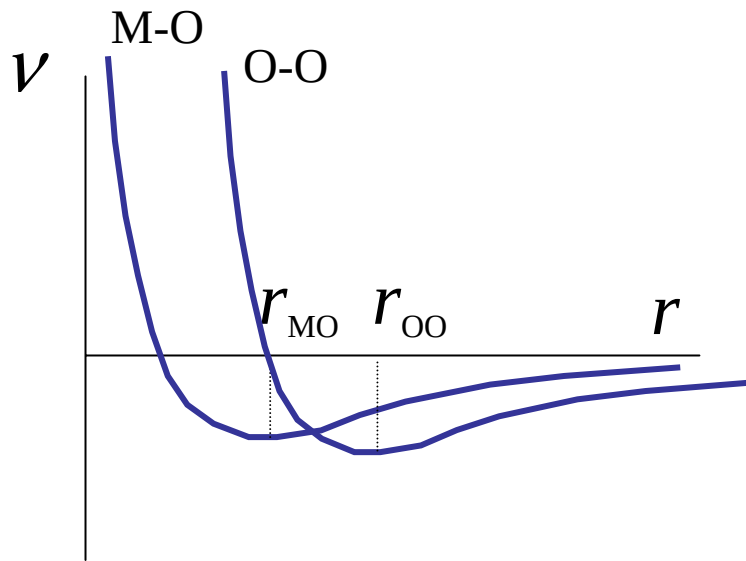
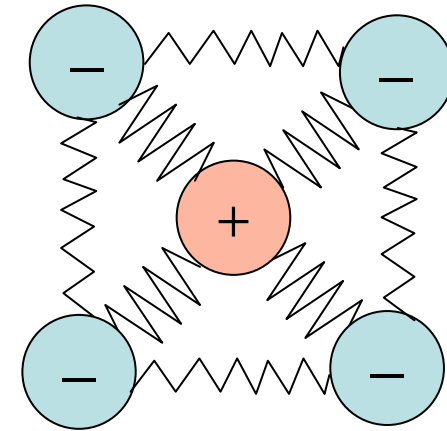


Static Lattice Energy Calculations

Empirical pair potentials

$$v_{ij} = \frac{q_i q_j}{r_{ij}} + \phi_{ij}(r)$$

$$E_{static} = 1/2 \sum_{i,j} v_{ij}$$



$$\frac{\partial E}{\partial x_i} = 0$$

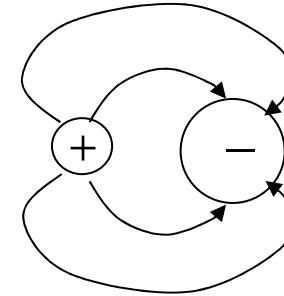
for all parameters x_i

THB potentials set for silicates

Sanders et al., 1984

Electrostatic potential

$$\frac{q_i q_j}{r}$$

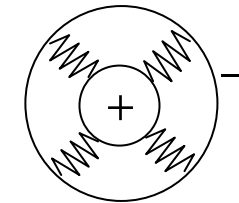


Buckingham potential

$$A \exp(-r/B) - Cr^{-6}$$

Core – shell interaction

$$1/2 K r^2$$

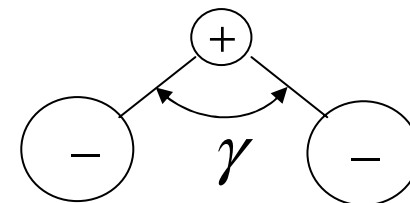


$$q_s = -2.848$$

$$q_c = +0.848$$

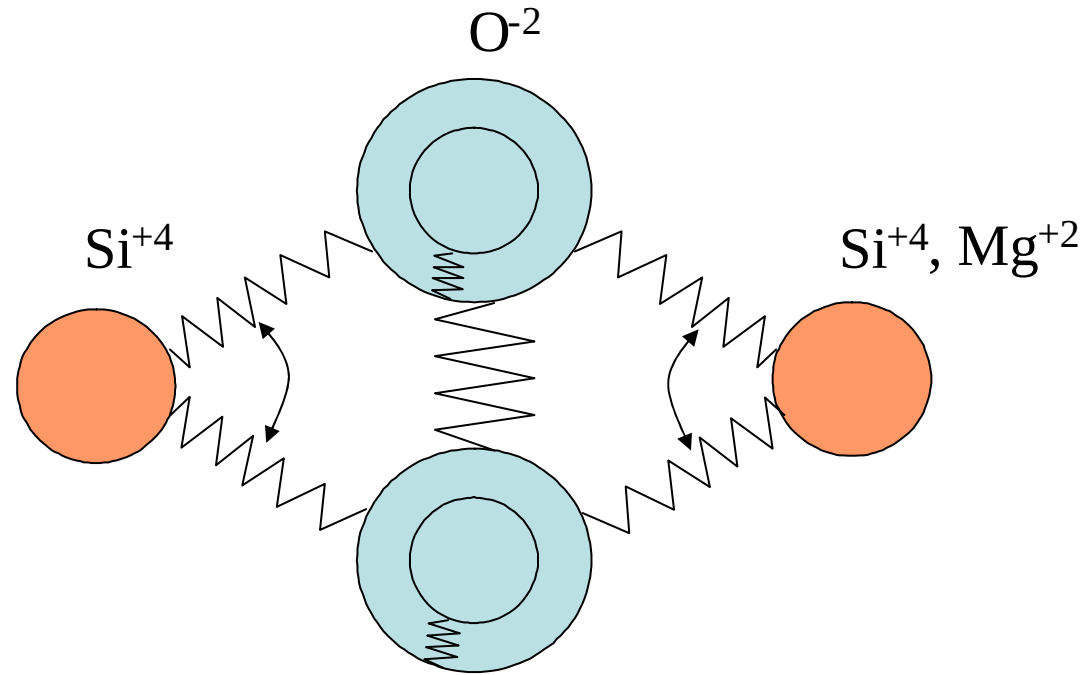
Bond-bending interaction

$$1/2 k (\gamma - \gamma_0)^2$$



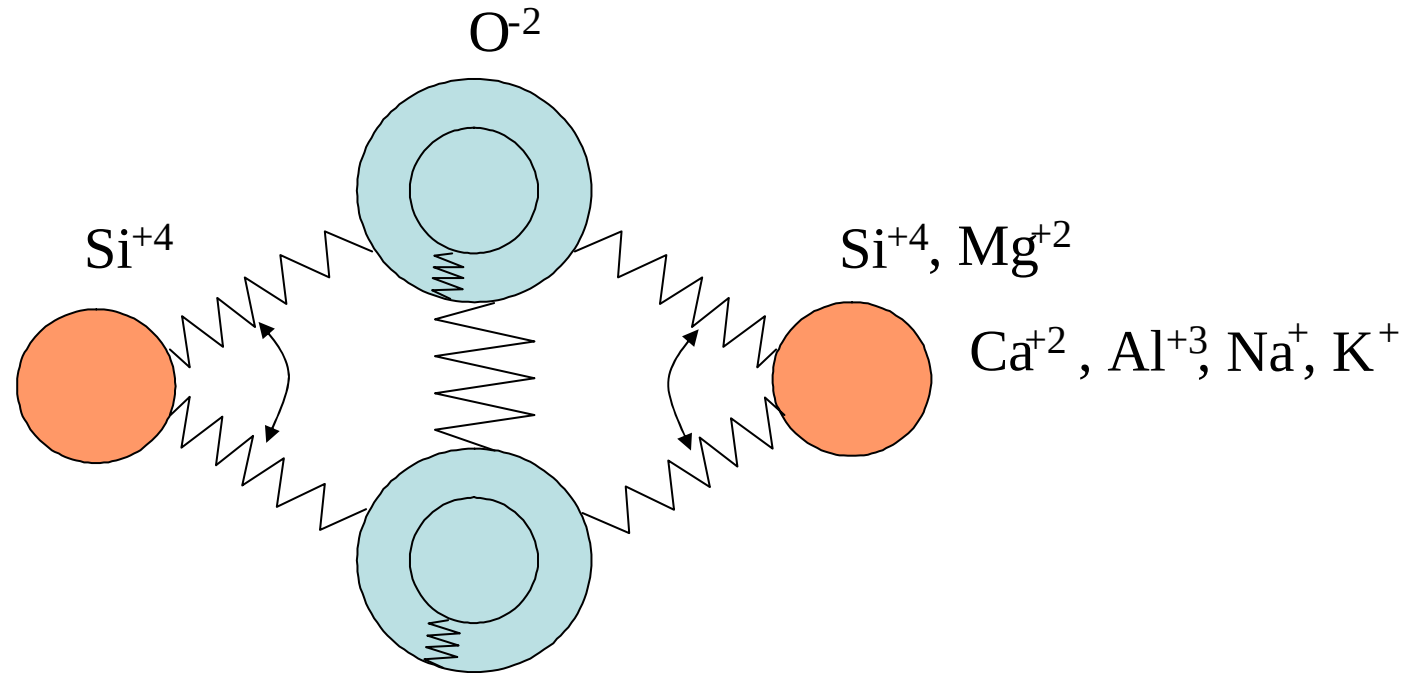
THB model for Mg-Si-O

Price et al., 1987

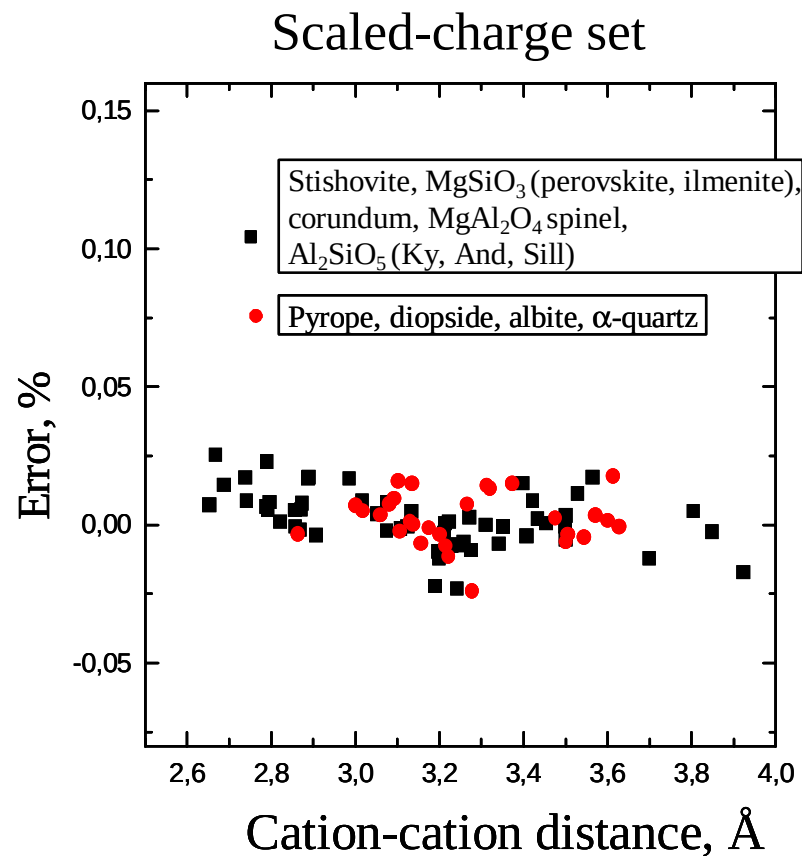
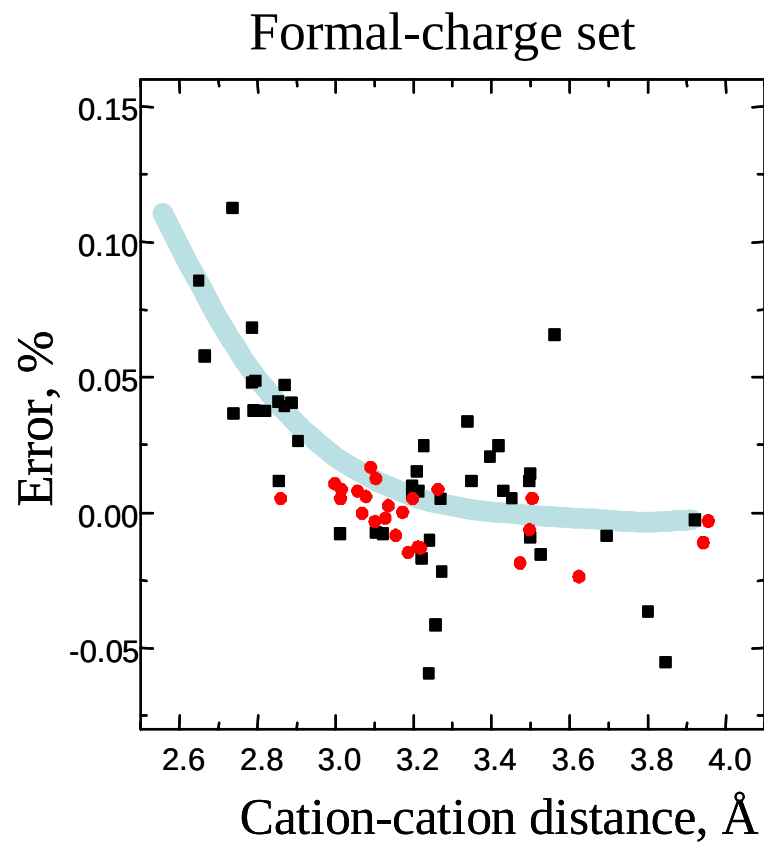


THB transferable set for aluminosilicates

Winkler et al., 1991



Scaling the charges permits to decrease the error



New scaled-charge potentials for oxides

Type	atom	atom	A(eV)	B(Å)	C(eV*Å ⁶)
Buckingham	Ca	O-shell	2895.68	0.2811	0
Buckingham	Mg	O-shell	1077.55	0.2899	0
Buckingham	Na	O-shell	30267.4	0.1997	0
Buckingham	K	O-shell	65164.8	0.2122	0
Buckingham	Al	O-shell	1226.70	0.2893	0
Buckingham	Si	O-shell	1096.42	0.2999	0
Buckingham	O-shell	O-shell	614.71	0.3016	27.07

Type	atom	atom	K(eV*Å ⁻²)
String	O-core	O-shell	54.70

Type	atom	atom	atom	k(eV*grad ⁻²)	g(grad)
Three-body	Si(4)	O-shell	O-shell	3.79	109.47
Three-body	Si(6)	O-shell	O-shell	3.77	90
Three-body	Al(4)	O-shell	O-shell	0.67	109.47
Three-body	Al(6)	O-shell	O-shell	1.89	90

Note: O core charge 0.7465; shell charge -2.4465 ,
the charges on cations are $0.85Z$.

Available also for Fe^{2+} , Fe^{3+} , Mn^{2+} , Ge, Zr, Ti, Y, P, F^- , C, OH^-

General Utility Lattice Program



Julian Gale

http://nanochemistry.curtin.edu.au/t2_julian.html

General Utility Lattice Program



Julian Gale

http://nanochemistry.curtin.edu.au/t2_julian.html

An example of GULP input-file

```
# Keywords:
conp prop opti phon
title
Calculation of standard entropy of pyrope
end
name pyrope
Temperature 298.15 K

cell
  11.492000  11.492000  11.492000  90.000000  90.000000  90.000000
fractional
Mg   core 0.0000000 0.2500000 0.1250000 2.00000000 1.00000 0.00000
Al2  core 0.0000000 0.0000000 0.0000000 3.00000000 1.00000 0.00000
Si1  core 0.0000000 0.2500000 0.3750000 4.00000000 1.00000 0.00000
O    core 0.0327999 0.0502000 0.6533999 0.84820000 1.00000 0.00000
O    shel 0.0327999 0.0502000 0.6533999 -2.8482000 1.00000 0.00000
space
I A 3 D
observables
elastic
  1 1  29.6200
elastic
  1 2  11.1100
elastic
  4 4   9.1600
bulk_modulus
  17.2800
shear_modulus
   9.2000
sdlc
  1 1  12.0000
end
buck
Mg   core 0  shel 1092.7076  0.327890  47.615886  0.0 12.00 1 1 1
Al   core 0  shel 1110.9213  0.323853  0.00000000  0.0 12.00 0 0 0
O    shel 0  shel 11540.311  0.089509  26.330266  0.0 12.00 0 0 0
Si   core 0  shel 1425.4929  0.319796  19.669306  0.0 12.00 0 0 0
spring
O    73.642708  0
three
Si1  core 0  shel 0  shel 1.8860  109.470000 &
      0.000 1.840 0.000 1.840 0.000 3.200 0 0
three
Al2  core 0  shel 0  shel 2.7146  90.000000 &
      0.000 2.200 0.000 2.200 0.000 3.200 1 0
shrink
4
dump every 1 pyrope.dump
```

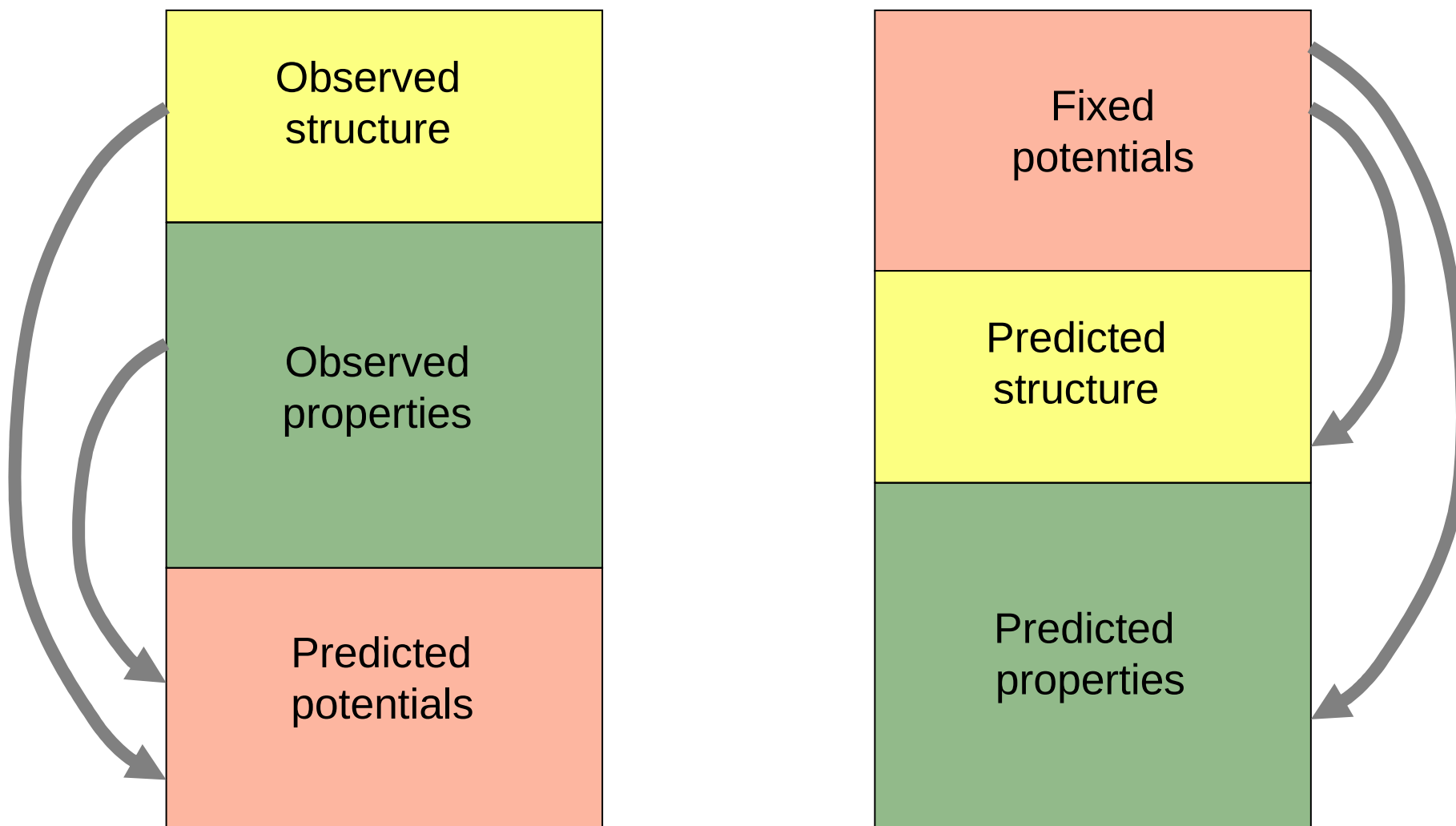
KEYWORDS

STRUCTURE

OBSERVABLES

POTENTIALS

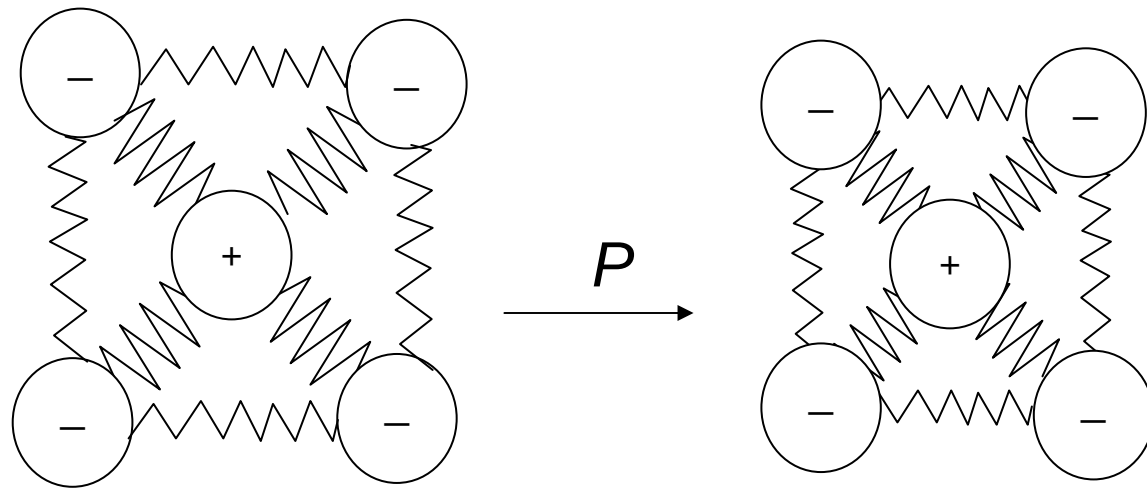
GULP can be used in two main ways:



Accuracy in the prediction of structure

	Cell-parameter	Experimental	Predicted	Difference (in percent)
Low albite	a	7.715	7.746	0.43
	b	7.437	7.499	0.83
	c	7.158	7.142	-0.22
	α	79.58	79.183	-0.5
	β	107.32	106.97	-0.33
	γ	64.92	65.26	0.53
	Volume	331.93	336.75	1.45
Microcline	a	7.913	7.930	0.21
	b	7.626	7.640	0.19
	c	7.222	7.205	-0.24
	α	76.32	76.44	0.16
	β	104.19	104.19	0
	γ	66.92	67.04	0.17
	Volume	360.65	362.05	0.39
Anorthite	a	8.180	8.223	0.53
	b	12.875	12.954	0.62
	c	14.172	14.137	-0.25
	α	93.13	92.97	-0.18
	β	115.89	116.08	0.17
	γ	91.24	90.53	-0.78
	Volume	1338.993	1349.918	0.82

Simulation of the effect of pressure

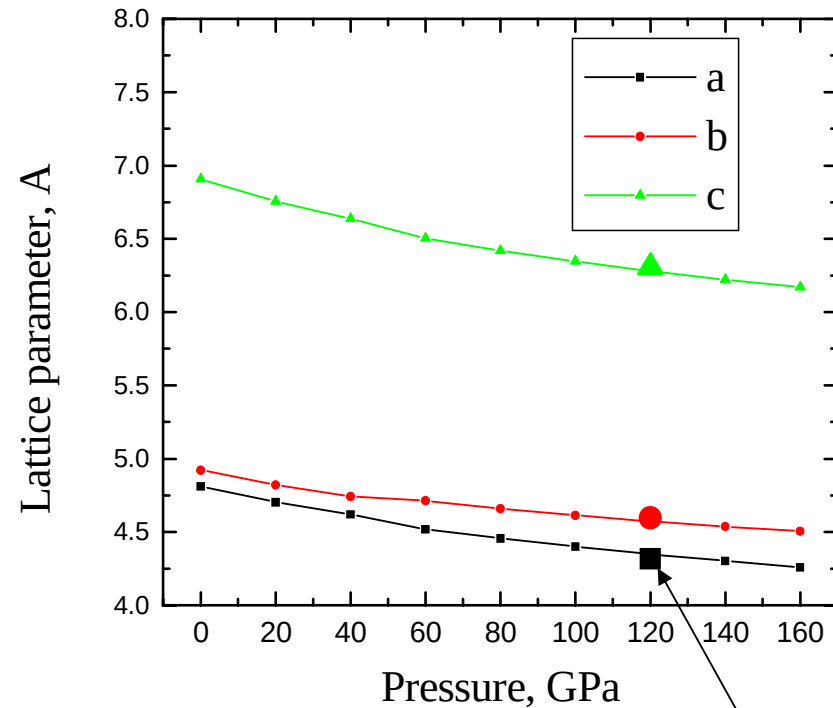
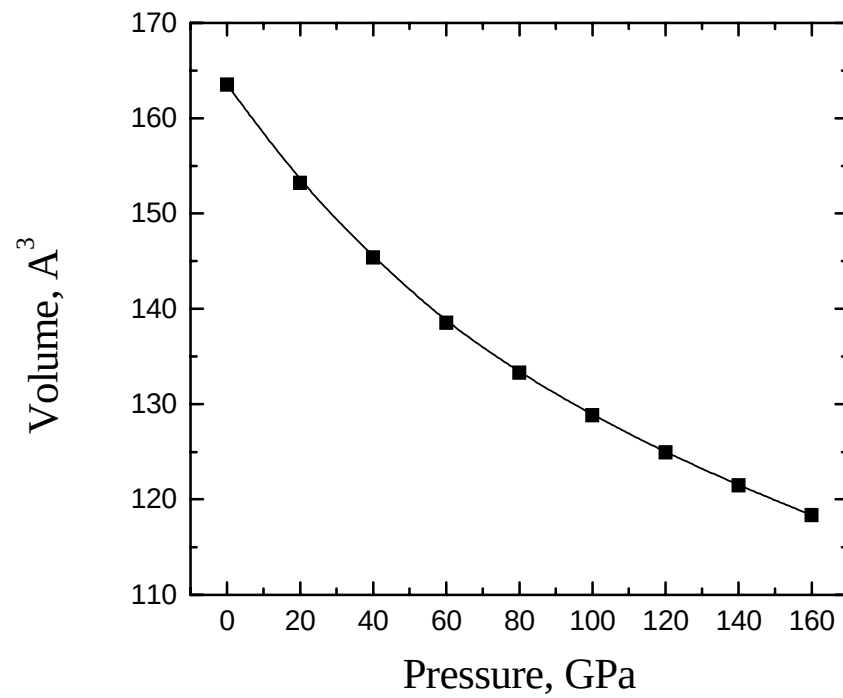


$$H = E + PV$$

Enthalpy

$$\frac{\partial H}{\partial x_i} = 0$$

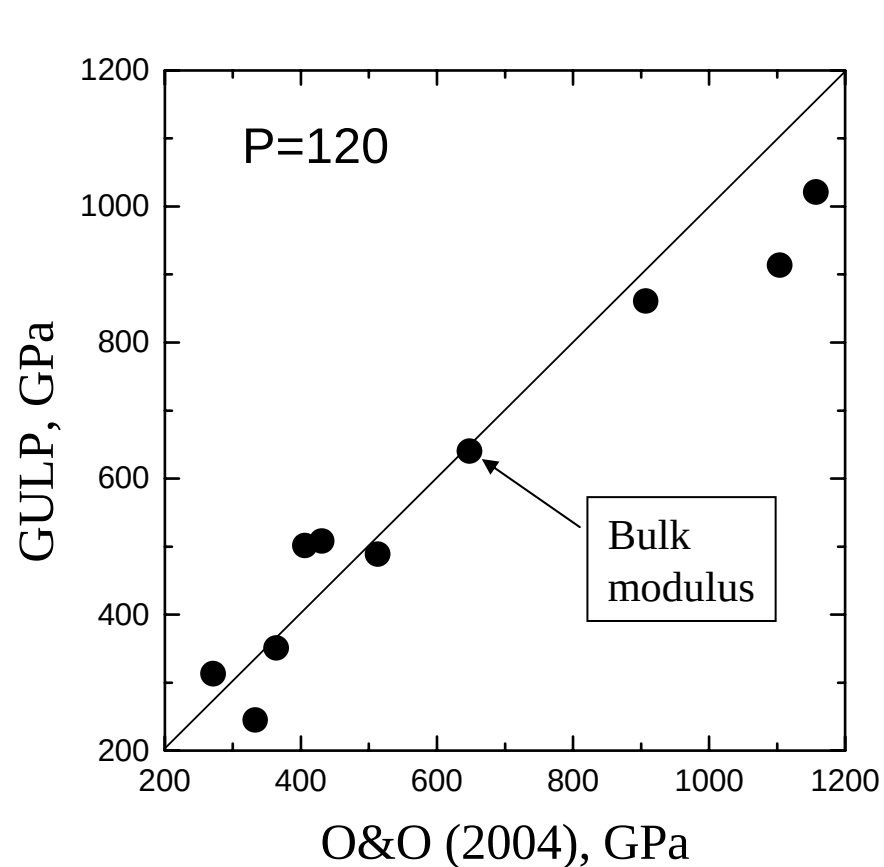
MgSiO₃ perovskite at high pressures



Large symbols: ab initio results
of Oganov & Ono (2004)

Elastic stiffness and lattice constants of perovskite at 120 GPa

GULP vs. DFT

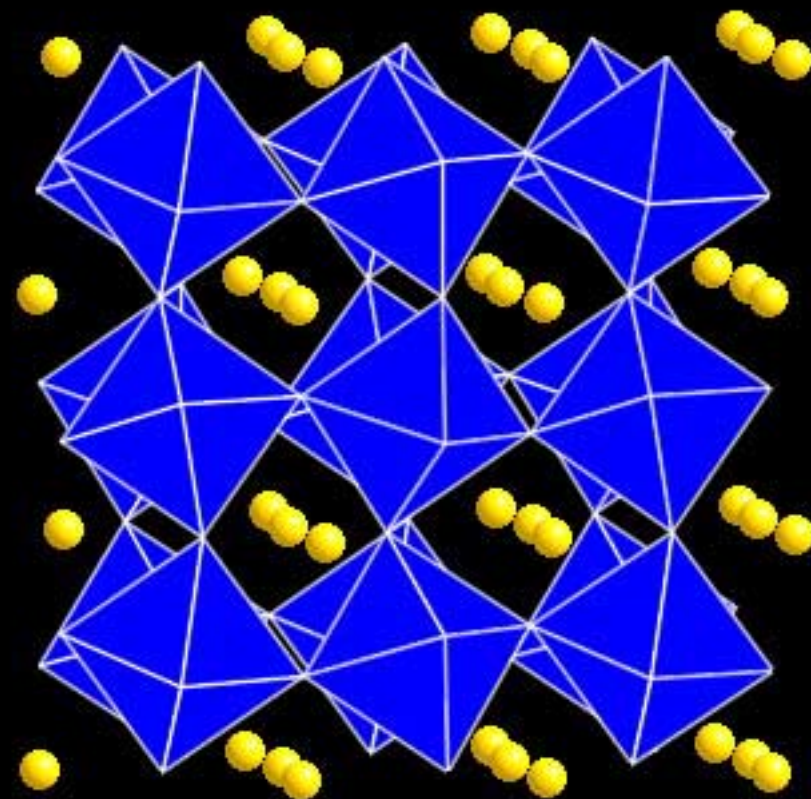


Oganov & Ono, 2004
ab initio

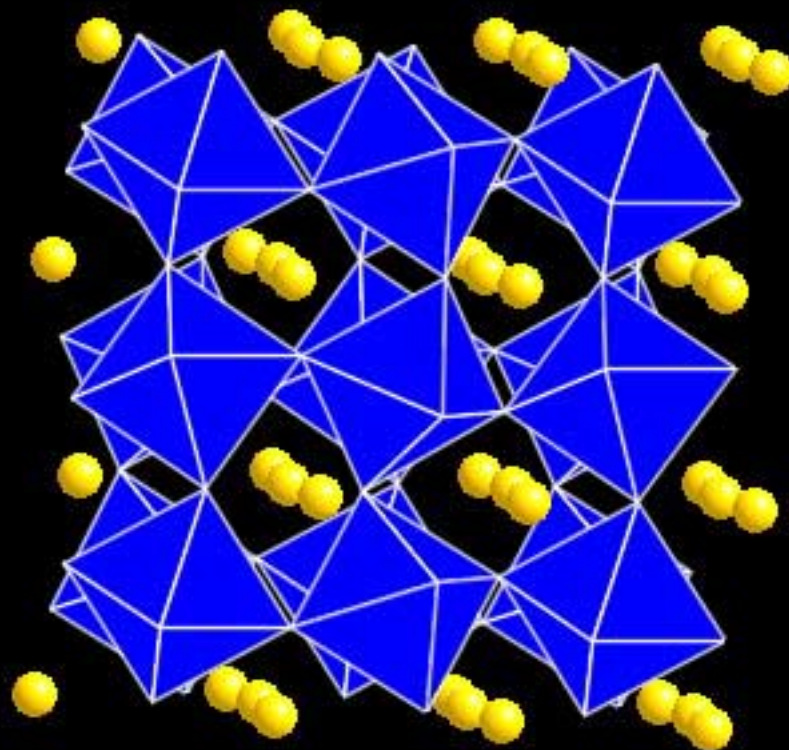
	O&O	GULP
<i>a</i>	4.318	4.348
<i>b</i>	4.595	4.574
<i>c</i>	6.305	6.281

$$K = V \frac{\partial^2 E}{\partial V^2}$$

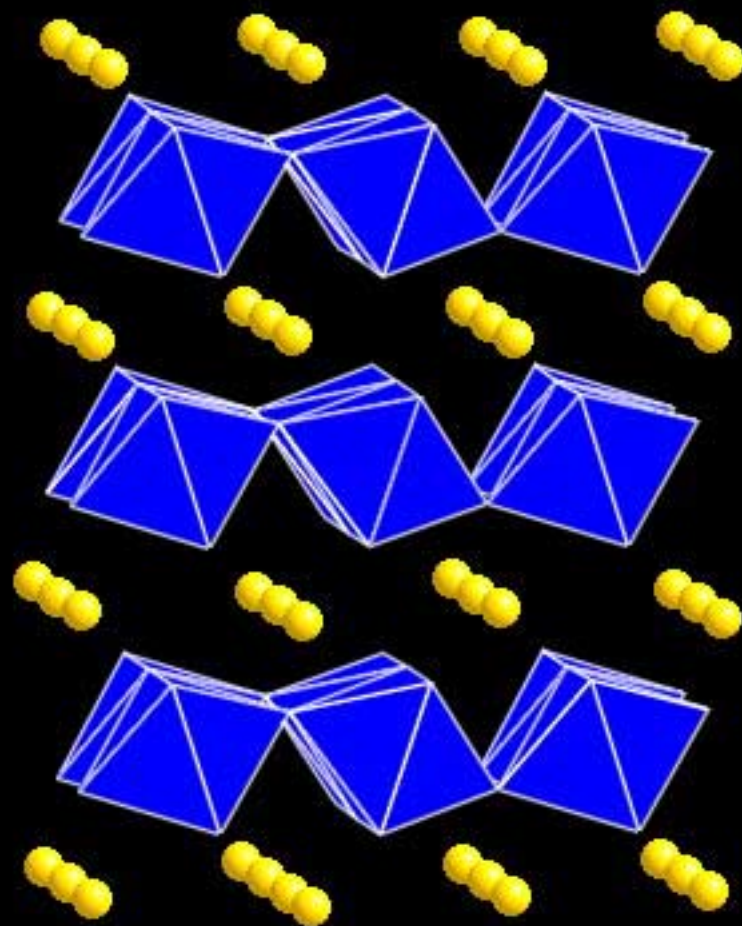
Perovskite at 0 GPa



Perovskite at 120 GPa

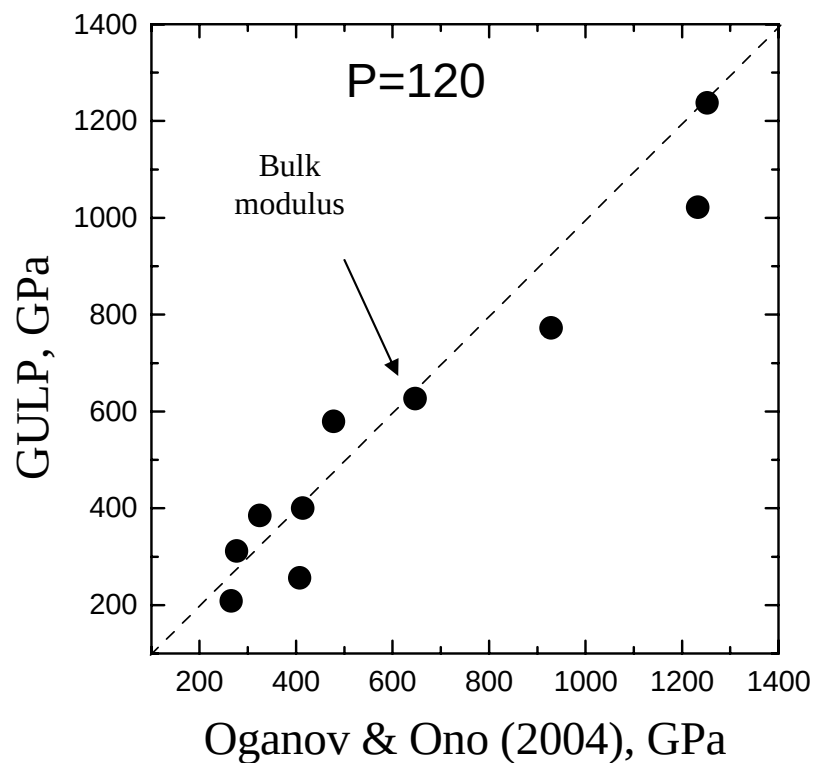


Post-perovskite at 120 GPa



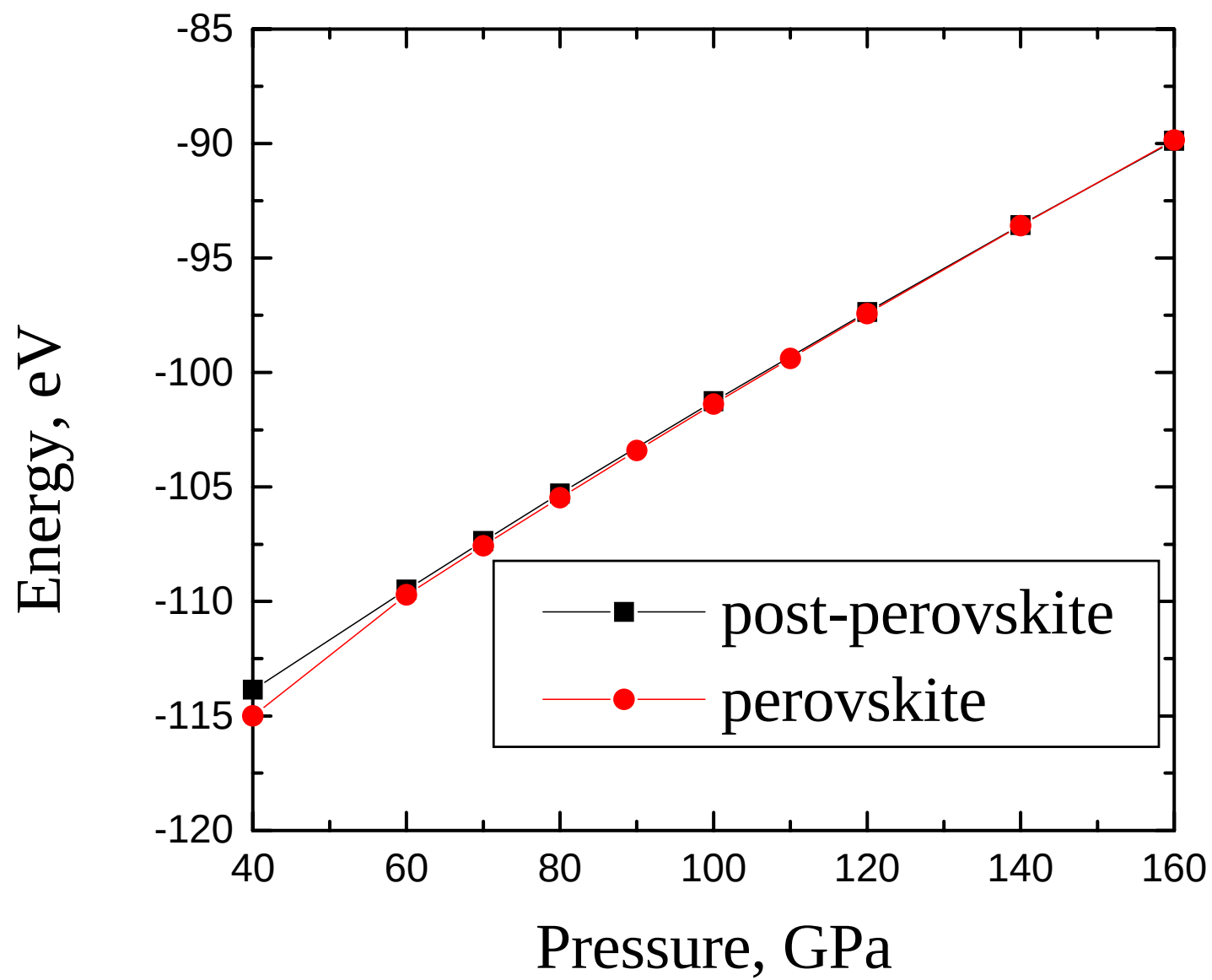
Oganov & Ono 2004

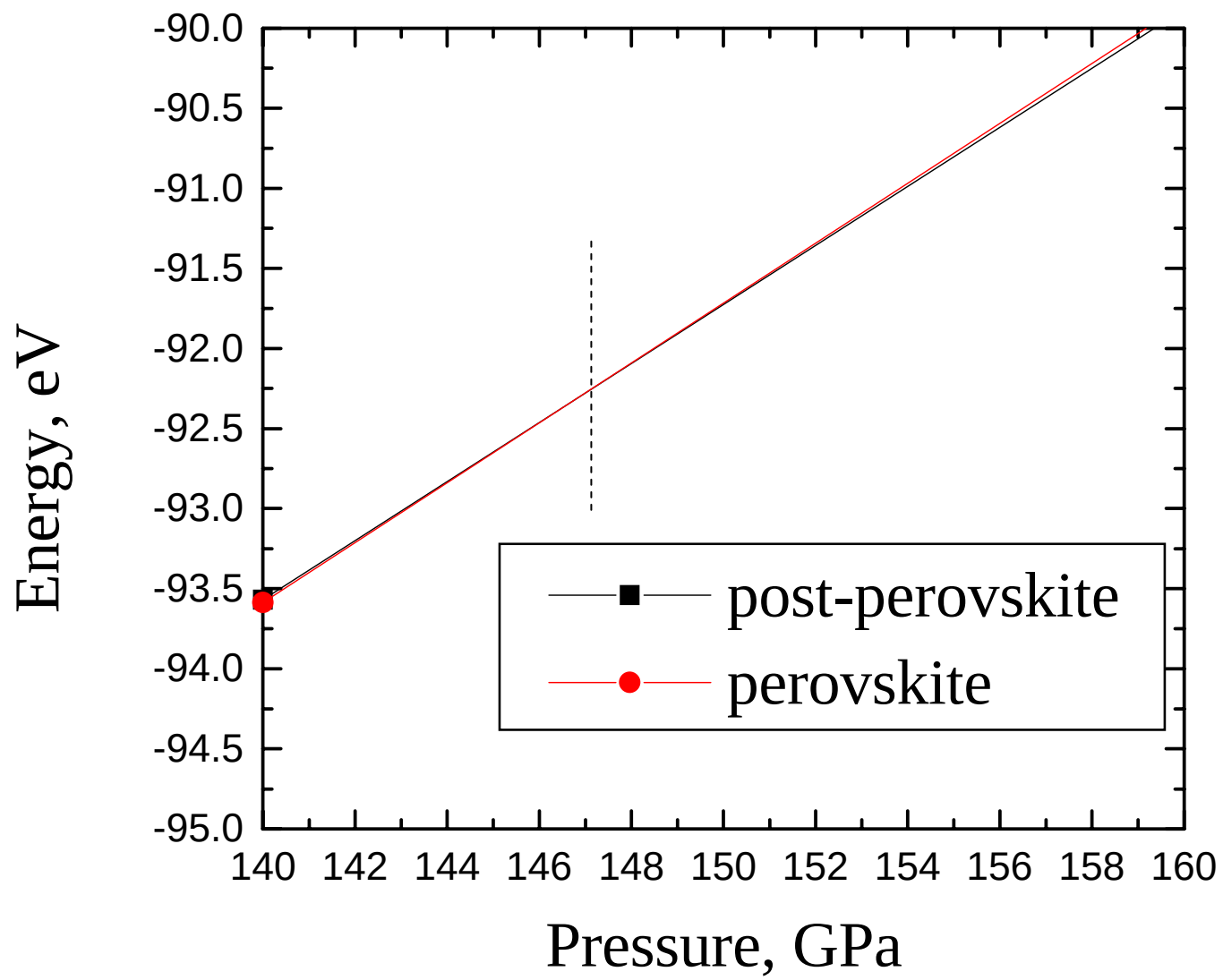
Elastic stiffness and lattice constants of post-perovskite at 120 GPa

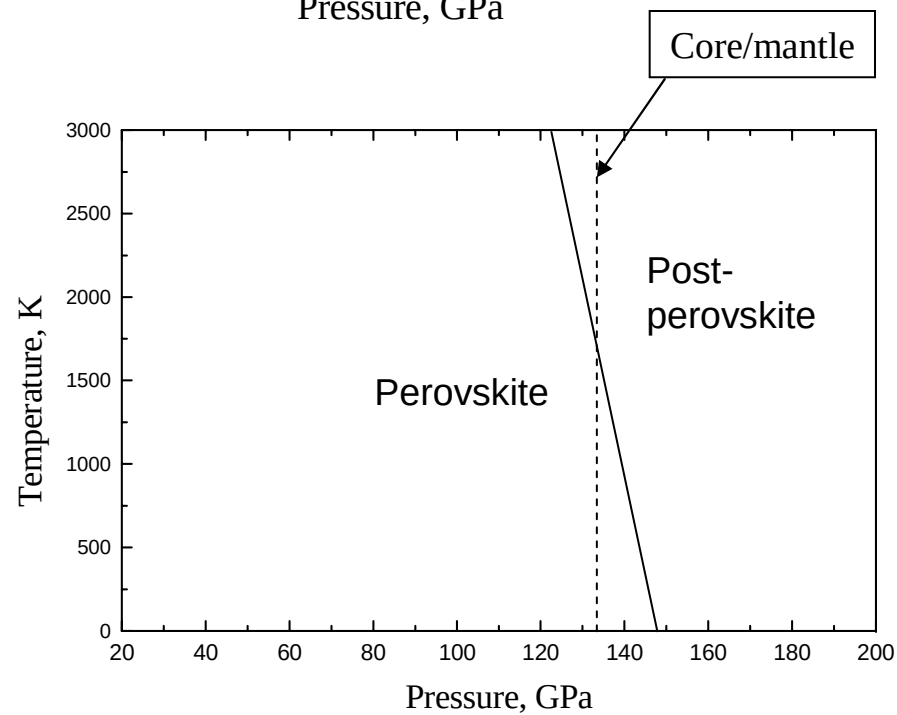
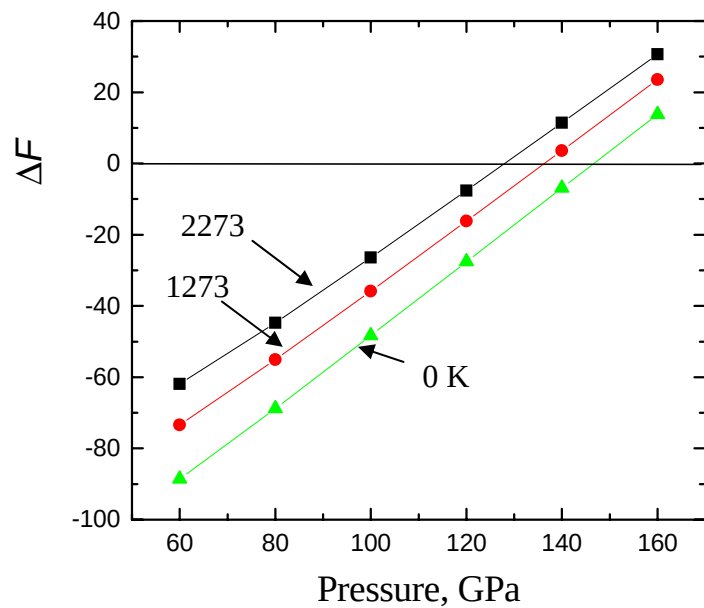
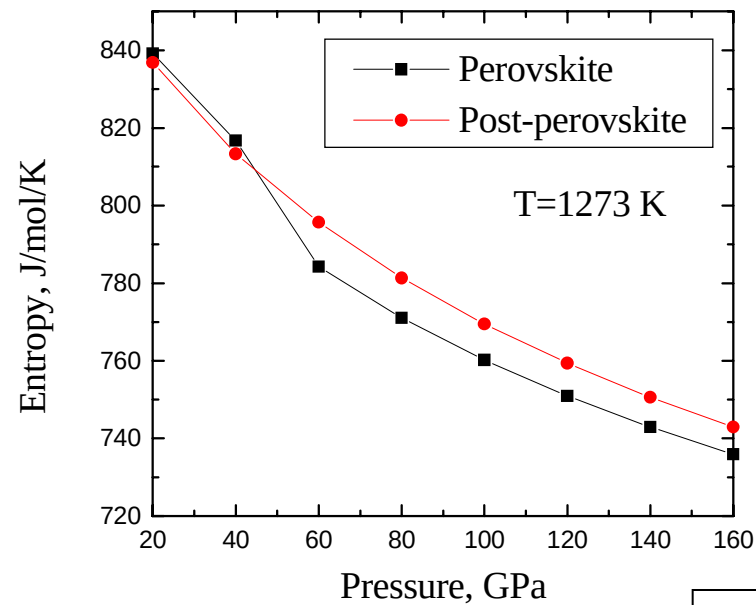
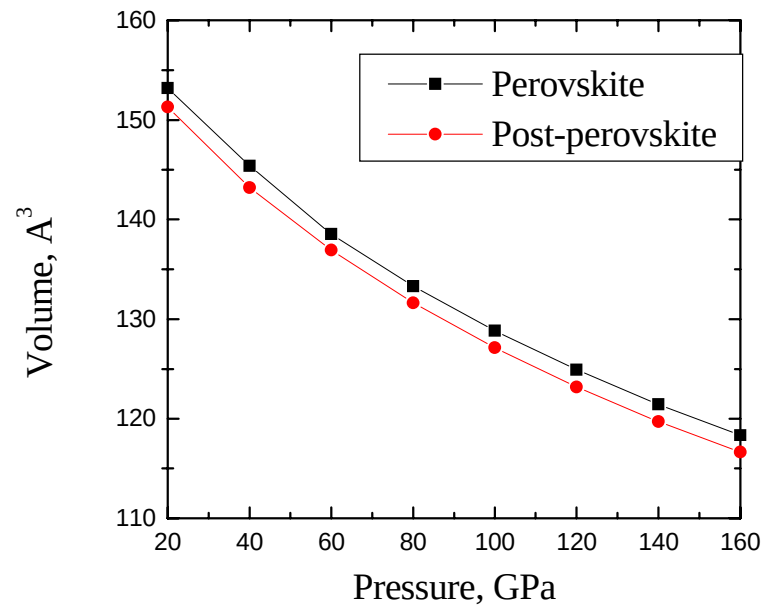


Oganov & Ono, 2004
ab initio

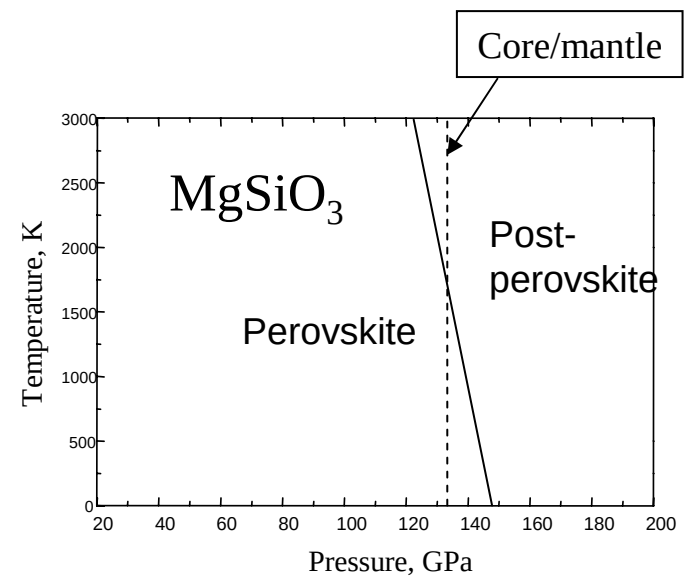
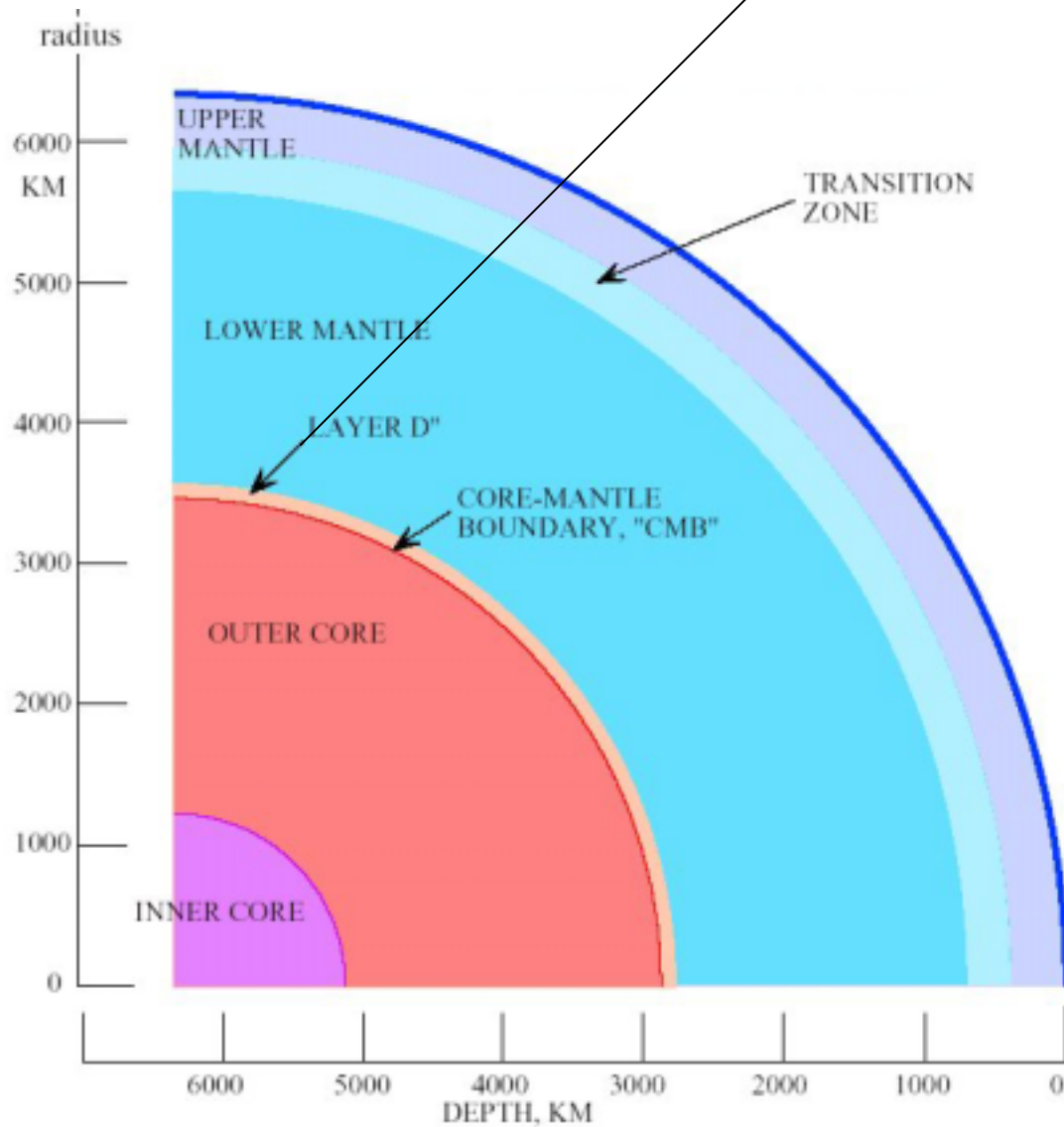
	O&O	GULP
<i>a</i>	2.474	2.492
<i>b</i>	8.121	8.159
<i>c</i>	6.138	6.058





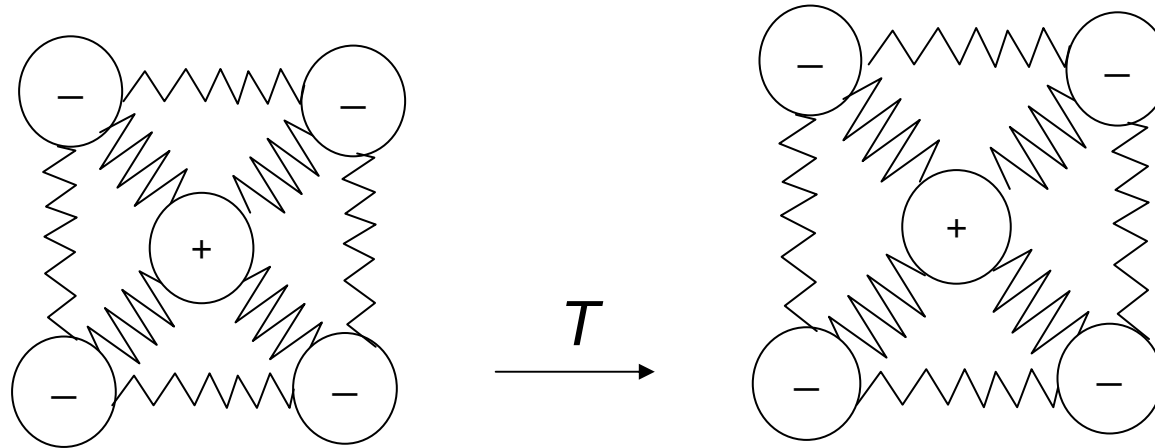


Perovskite/post-perovskite transition



It is important to determine the right slope!

Simulation of the effect of temperature

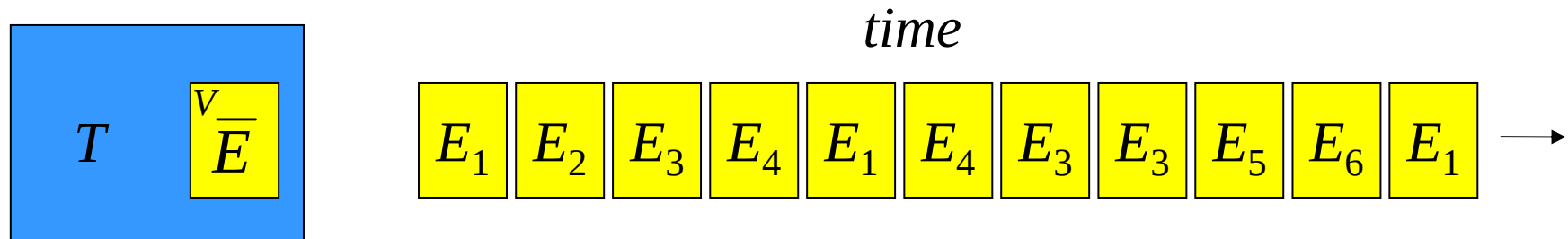


$$F = E_{static} + E_{vib} - TS_{vib}$$

$$\frac{\partial F}{\partial x_i} = 0$$

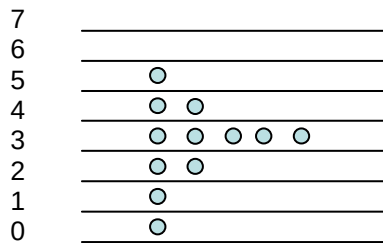
The system at T,V: The Boltzmann distribution

The energy of a thermally equilibrated system will fluctuate around the average energy $\overline{E}$



All sufficiently long successions will be statistically the same

A typical succession of microstates has certain specific properties, namely each configuration occurs with the Boltzmann probability



$$p_i = \frac{e^{-E_i/kT}}{\sum_i e^{-E_i/kT}}$$

We collect the systems and make an isolated ensemble

E_1	E_2	E_3	E_4
E_1	E_4	E_3	E_3
E_5	E_6	E_1	E_7

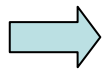
The energy of the ensemble is a sum of E_i

There will be many microstates with the same total energy

The equilibrium of the ensemble corresponds to the maximum number of microstates

The main postulate of Statistical Mechanics:

all microstates in an isolated system have the same probability



The state with the maximum of microstates has the highest probability

Boltzmann distribution

The number of distinguishable microstates in the ensemble is

$$W(M) = \frac{M!}{\prod_i M_i!}$$

The entropy of an isolated ensemble of M systems can be defined

$$S(M) = k \ln \frac{M!}{\prod_i M_i!} = -kM \sum_i p_i \ln p_i$$

Entropy per
1 system:

$$S = S(M) / M = -k \sum_i p_i \ln p_i$$

We can either maximize $W(M)$

$$W(M) = \frac{M!}{\prod_i M_i!} \quad \text{with respect to } M_i$$

or maximize the entropy

$$S = S(M) / M = -k \sum_i p_i \ln p_i \quad \text{with respect to } p_i$$

The set of p_i , which brings the maximum to the entropy of the thermally equilibrated system, is the Boltzmann distribution

The constancy of the average energy gives a constraint to the maximization procedure

$$\overline{E}(M) = M\overline{E} = M \sum_i E_i p_i$$

The normalization of the probabilities gives another constraint

$$\sum_i p_i = 1$$

The maximization of the entropy with the above constraints is equivalent to the maximization of the function

the undetermined Lagrange multipliers

$$S^* = -\sum_i p_i \ln p_i - \alpha \sum_i p_i - \beta \sum_i p_i E_i$$


without constraints

The condition of the maximum is that all partial derivatives are zero:

$$0 = \frac{\partial}{\partial p_i} \left(- \sum_i p_i \ln p_i - \alpha \sum_i p_i - \beta \sum_i p_i E_i \right)$$

$$0 = \frac{\partial}{\partial p_i} \left(- \sum_i (p_i \ln p_i + \alpha p_i + \beta E_i p_i) \right)$$

$$0 = p_i \frac{1}{p_i} + \ln p_i + \alpha + \beta E_i$$

$$p_i = e^{-(1+\alpha)} e^{-\beta E_i}$$

The physical meaning of the α parameter can be found out from the normalization condition

$$\sum_i p_i = 1 \quad \Rightarrow \quad \sum_i p_i = e^{-(1+\alpha)} \sum_i e^{-\beta E_i} = 1$$

$$\Rightarrow \quad e^{-(1+\alpha)} = \frac{1}{\sum_i e^{-\beta E_i}} = \frac{1}{Z}$$

The partition function
(Zustandssumme)

$$\Rightarrow \quad p_i = \frac{e^{-\beta E_i}}{\sum_i e^{-\beta E_i}}$$

The physical meaning of the β parameter can be found by substituting the Boltzmann probabilities back into the entropy equation

$$p_i = \frac{e^{-E_i\beta}}{\sum_i e^{-E_i\beta}} \implies S = -k \sum_i p_i \ln p_i \implies S = k\beta\bar{E} + k \ln Z$$

$$\implies \frac{\partial S}{\partial \bar{E}} = k\beta$$

In classical thermodynamics it is proved that $\frac{\partial S}{\partial \bar{E}} = \frac{1}{T}$.

$$\text{Thus } \beta = \frac{1}{kT}.$$

One can also see that

$$S = \frac{1}{T} \bar{E} + k \ln Z \implies -kT \ln Z = \bar{E} - TS$$

which compares with the classical result: $F = \bar{E} - TS$

The Boltzmann distribution law takes the final form

$$p_i = \frac{e^{-E_i/kT}}{\sum_i e^{-E_i/kT}}$$

when E_i decreases,
 p_i increases

high T tends,
to equalize p_i

The following two statements are equivalent:

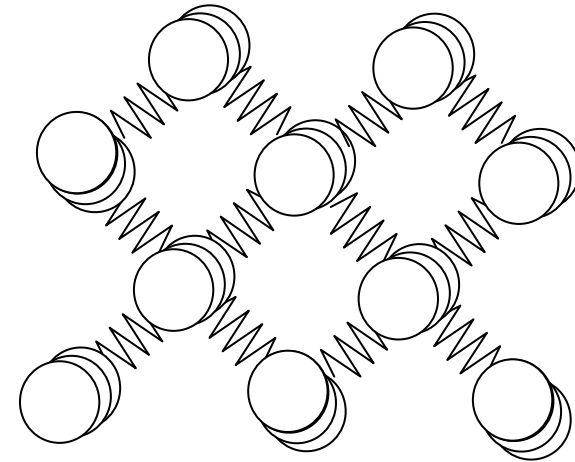
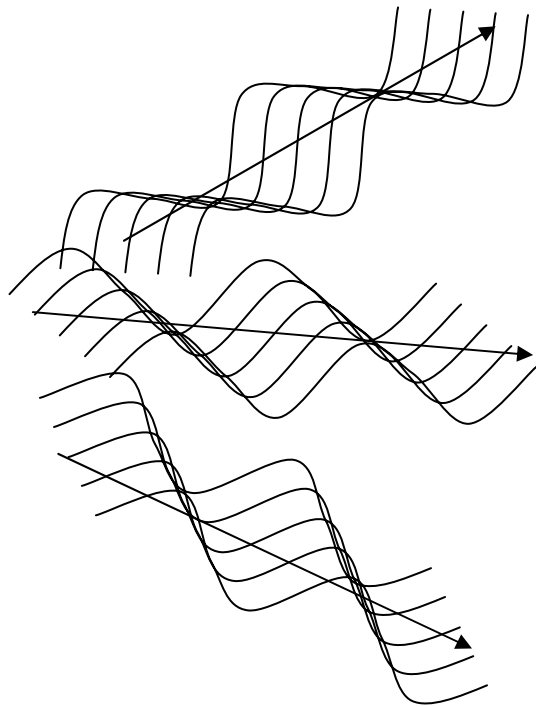
1) The free energy function achieves the minimum

$$-kT \ln Z = F = \bar{E} - TS$$

2) Energy states are distributed according to the
Boltzmann law

Lattice dynamics

The motions of atoms can be described as a superposition of $3mN$ independent lattice vibrations (modes)



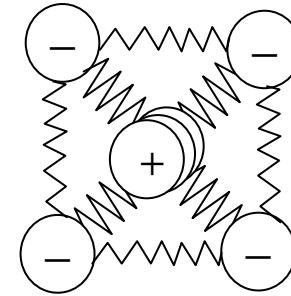
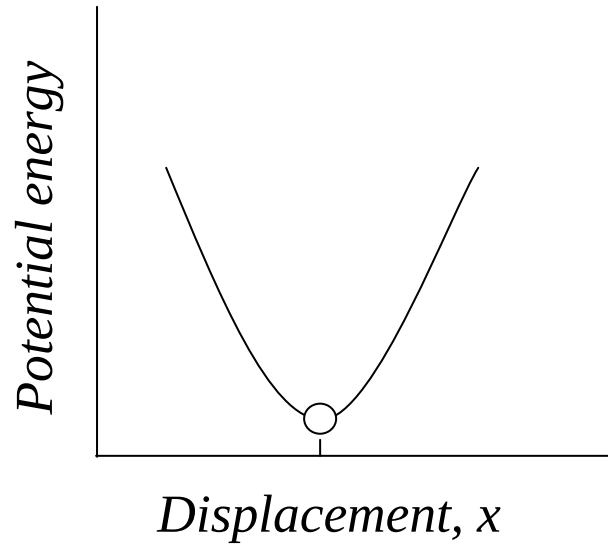
There are $3m$ branches for each k -vector and there are N k -vectors possible

$$E_{vib} = \sum_{\substack{\text{all modes} \\ k, \nu}} \bar{\epsilon}(k, \nu),$$

$$1 \leq k \leq N, \quad 1 \leq \nu \leq 3m$$

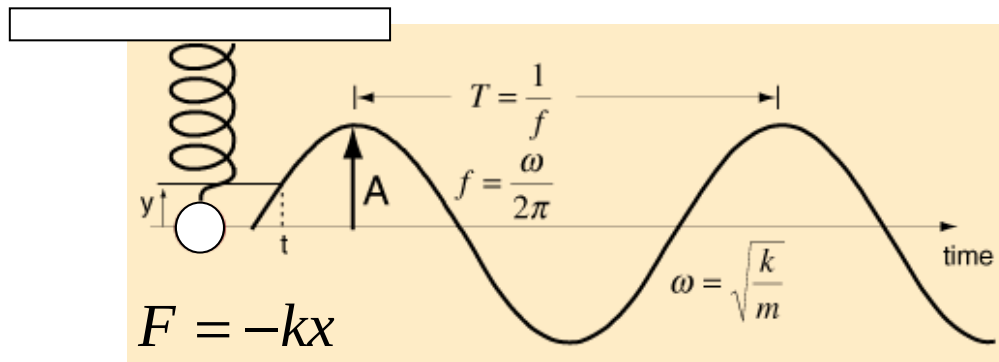
Each mode behaves as an independent oscillator and its energy is quantized

Vibrational energy at $T > 0$



Energy is a quadratic function of displacement

Linear harmonic oscillator



Motion is harmonic

If we know force constants,
we can calculate
frequencies
of lattice vibrations

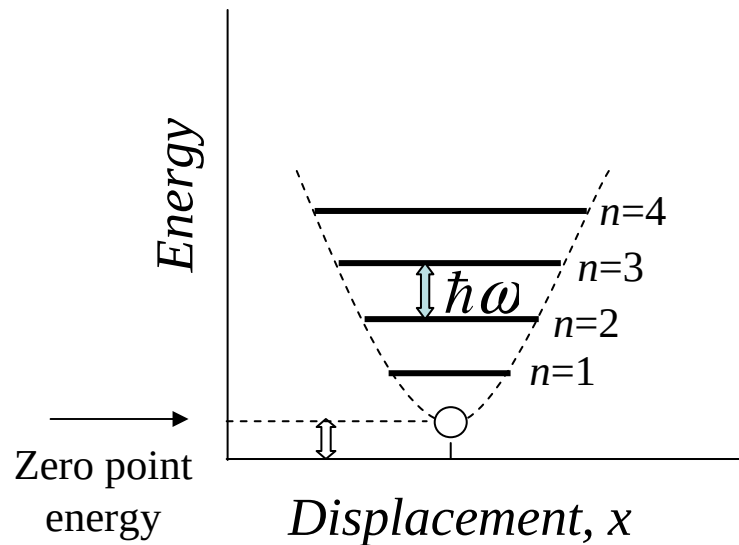
Einstein model

=

Born quantisation

+

Boltzmann
distribution



The energy of an oscillator depends on the frequency and excitation level

$$\varepsilon_n = \hbar\omega(1/2 + n)$$

The average energy is $\bar{\varepsilon} = \sum_n \varepsilon_n p_n$

where

$$p_n = \frac{e^{-\varepsilon_n/kT}}{\sum_i e^{-\varepsilon_i/kT}}$$

Therefore the energy of each oscillator is

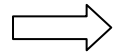
$$\bar{\varepsilon}_n = \hbar\omega \left(1/2 + \frac{1}{e^{\hbar\omega/kT} - 1} \right) = \hbar\omega(1/2 + \bar{n})$$

Since there are $3mN$ linear harmonic oscillators

$$E = 3mN\bar{\varepsilon}_n$$

Einstein

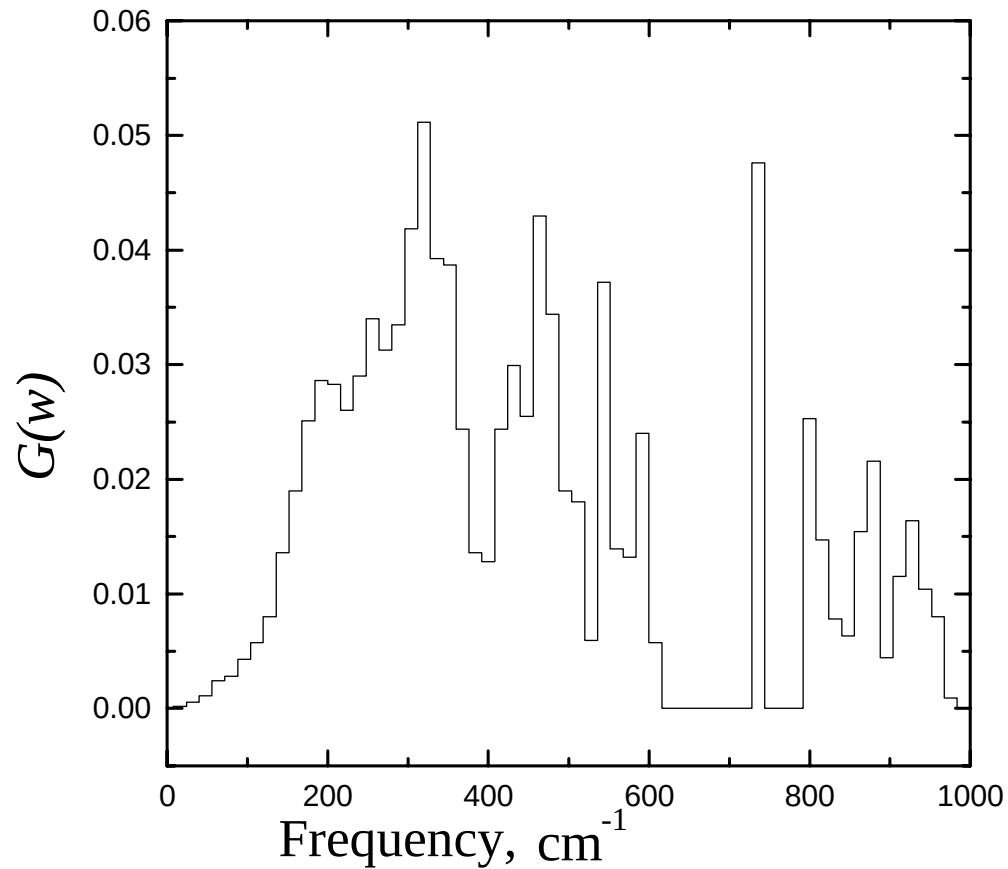
$$E = 3Nm\hbar\omega(1/2 + \bar{n})$$



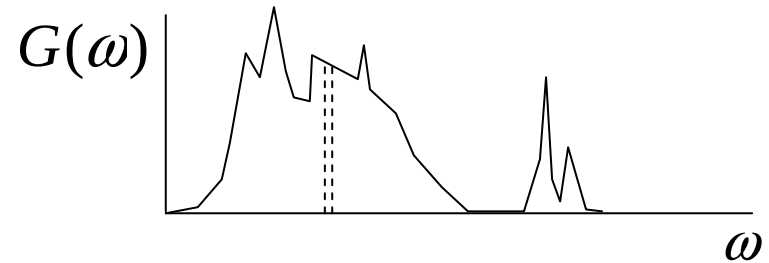
Born

$$E_{vib} = \int_0^{\omega_{max}} \hbar\omega(1/2 + \bar{n}(\omega, T))G(\omega)d\omega$$

The phonon density of states in forsterite



The thermodynamic output in GULP



$G(\omega)d\omega$ is the number
of modes in the interval
 $\omega, \omega+d\omega$

$$\int_0^{\omega_{\max}} G(\omega) d\omega = 3mN$$

$$E_{\text{vib}} = \int_0^{\omega_{\max}} \hbar\omega \left(\frac{1}{2} + \bar{n}(\omega, T) \right) G(\omega) d\omega$$

$$F = E_{\text{stat}} + \frac{1}{2} \int_0^{\omega_{\max}} \hbar\omega G(\omega) d\omega + k_{\text{B}}T \int_0^{\omega_{\max}} \ln(1 - \exp(-\hbar\omega / k_{\text{B}}T)) G(\omega) d\omega$$

Heat capacity

$$C_V = \left. \frac{\partial E}{\partial T} \right|_V$$

Vibrational entropy

$$S = - \left. \frac{\partial F}{\partial T} \right|_V$$

An example of a GULP run

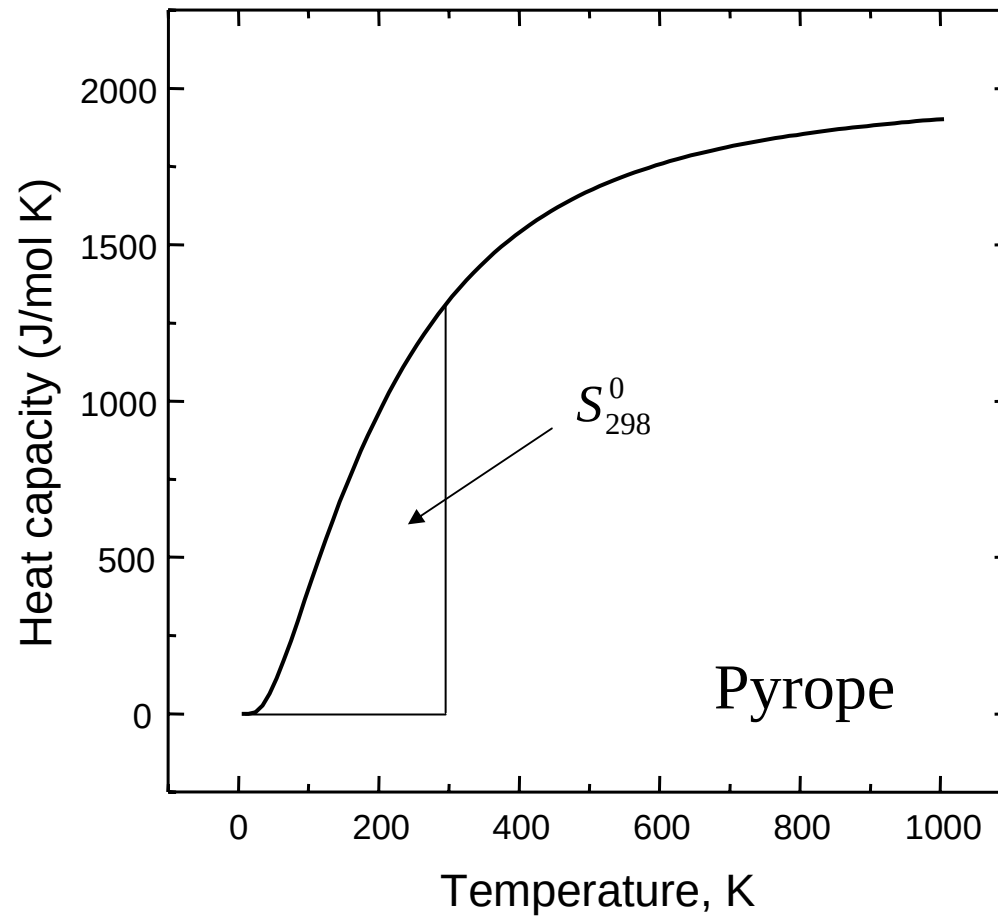
\$ gulp < pyrope.dat > pyrope.out

```
# Keywords:
conp prop opti phon
title
Calculation of standard entropy of pyrope
end
name pyrope
Temperature 298.15 K
cell
  11.492000  11.492000  11.492000  90.000000  90.000000  90.000000
fractional
Mg   core 0.0000000 0.2500000 0.1250000 2.000000000 1.00000 0.00000
Al2  core 0.0000000 0.0000000 0.0000000 3.000000000 1.00000 0.00000
Si1  core 0.0000000 0.2500000 0.3750000 4.000000000 1.00000 0.00000
O    core 0.0327999 0.0502000 0.6533999 0.84820000 1.00000 0.00000
O    shel 0.0327999 0.0502000 0.6533999 -2.8482000 1.00000 0.00000
space
I A 3 D
observables
elastic
  1 1 29.6200
elastic
  1 2 11.1100
elastic
  4 4 9.1600
bulk_modulus
  17.2800
shear_modulus
  9.2000
sdlc
  1 1 12.0000
end
buck
Mg   core 0  shel 1092.7076  0.327890  47.615886  0.0 12.00 1 1 1
Al   core 0  shel 1110.9213  0.323853  0.00000000 0.0 12.00 0 0 0
O    shel 0  shel 11540.311  0.089509  26.330266  0.0 12.00 0 0 0
Si   core 0  shel 1425.4929  0.319796  19.669306  0.0 12.00 0 0 0
spring
O    73.642708  0
three
Si1  core 0  shel 0  shel 1.8860  109.470000 &
     0.000 1.840 0.000 1.840 0.000 3.200 0 0
three
Al2  core 0  shel 0  shel 2.7146  90.000000 &
     0.000 2.200 0.000 2.200 0.000 3.200 1 0
shrink
4
dump every 1 pyrope.dump
```

Calculation of the standard
entropy of pyrope



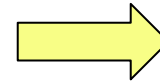
The heat capacity derived from the GULP output



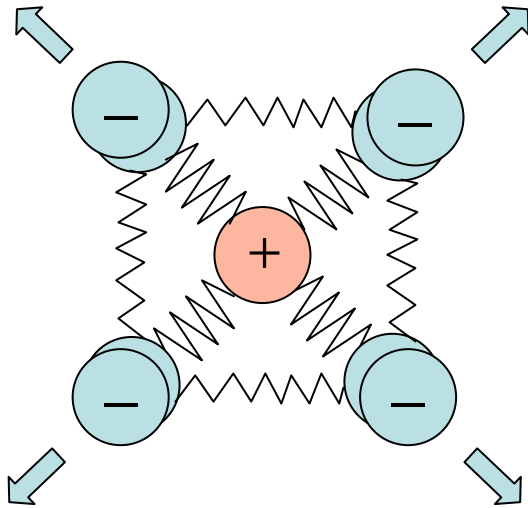
$$S^0_{298} \approx \int_0^{298.15} \frac{C_V(T)}{T} dT$$

Calculation of thermal expansion

An increase in volume leads to a decrease of vibrational frequencies. This, in turn, leads to an increase in the vibrational entropy



Thermal expansion is favoured

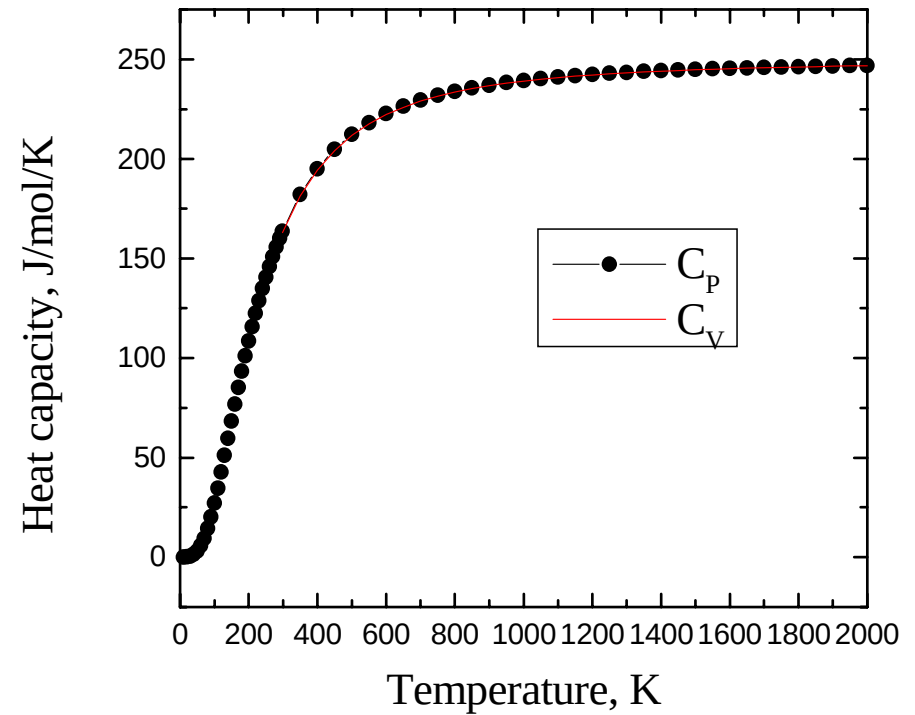
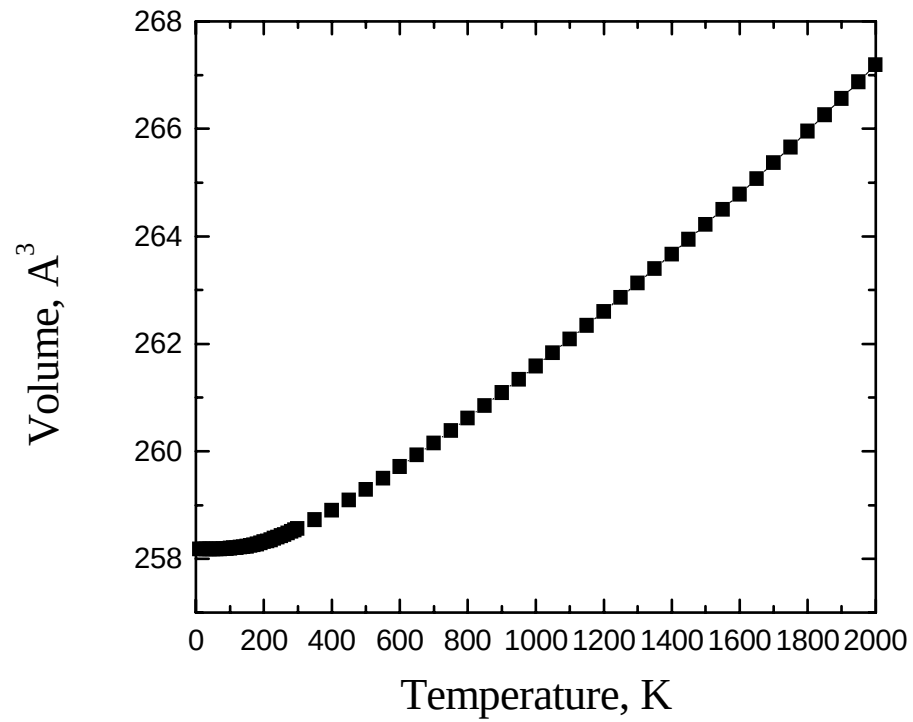


$$F = E_{static} + E_{vib} - TS_{vib}$$

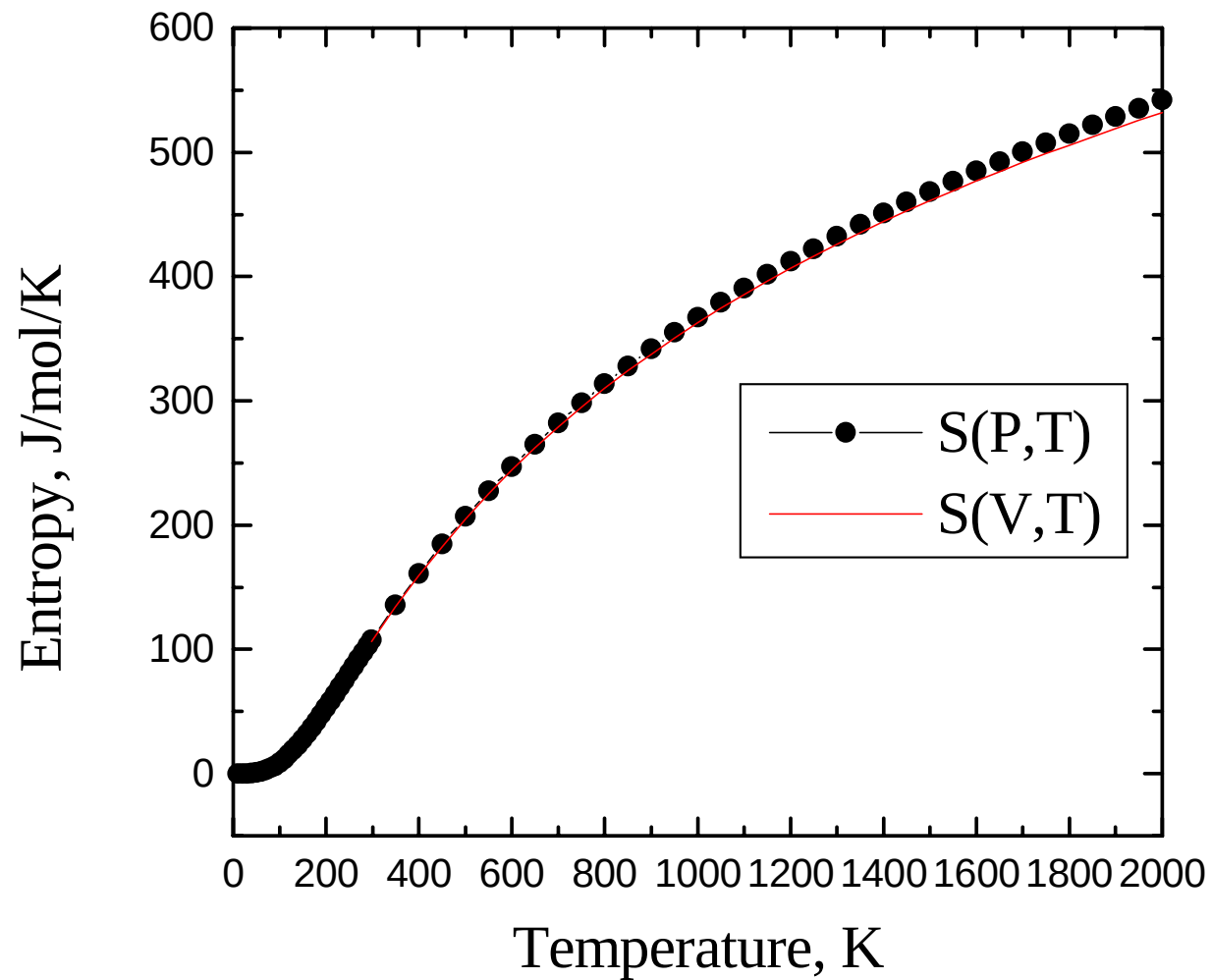
$$\frac{\partial F}{\partial x_i} = 0$$

Keyword: zsis

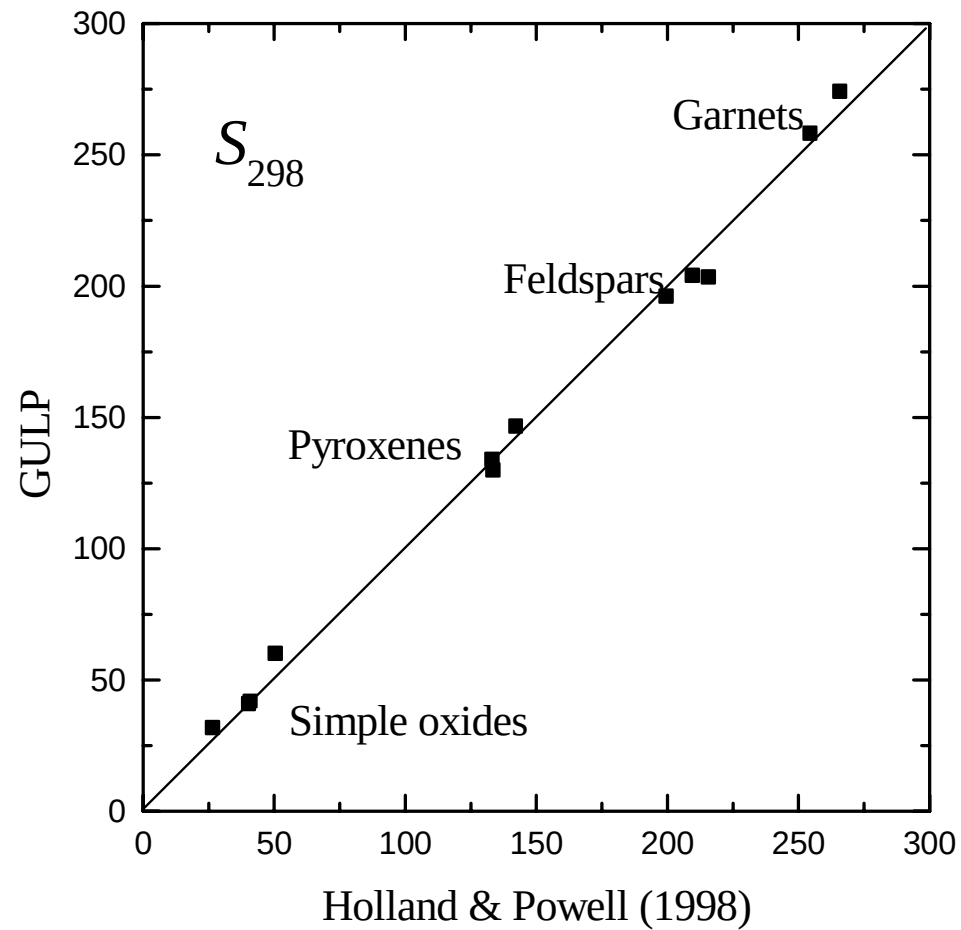
Corundum in quasi-harmonic approximation

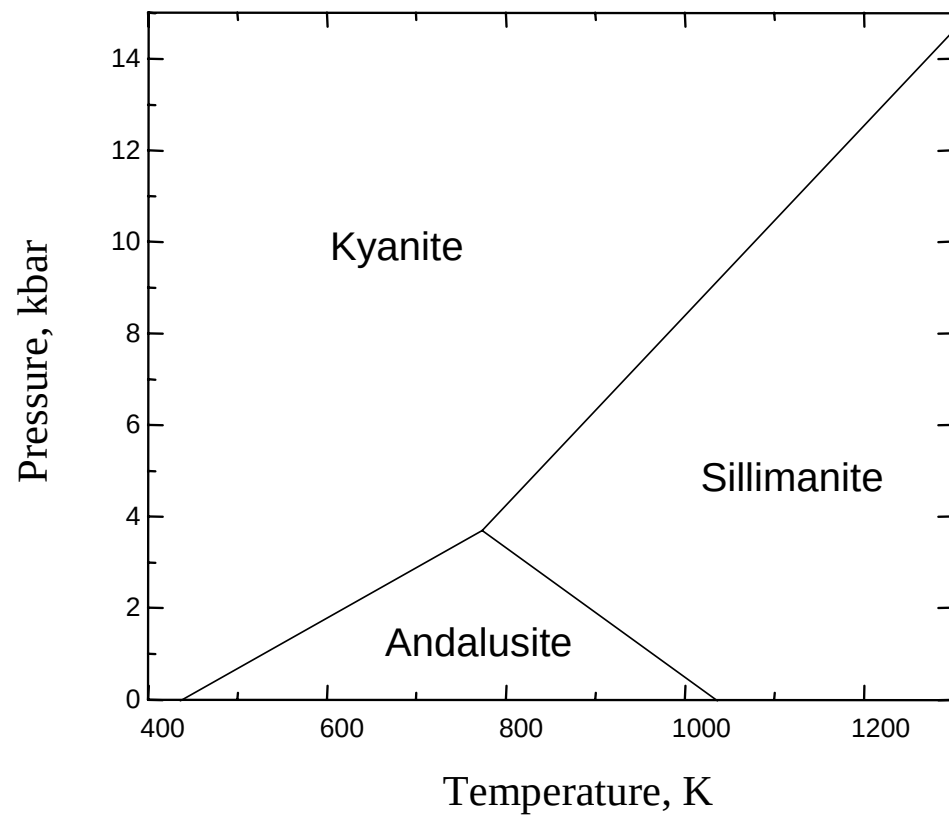


Entropy of corundum in quasi-harmonic approximation

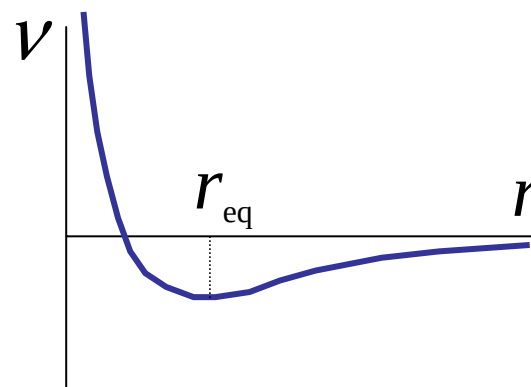


Correlation between predicted and experimental entropies

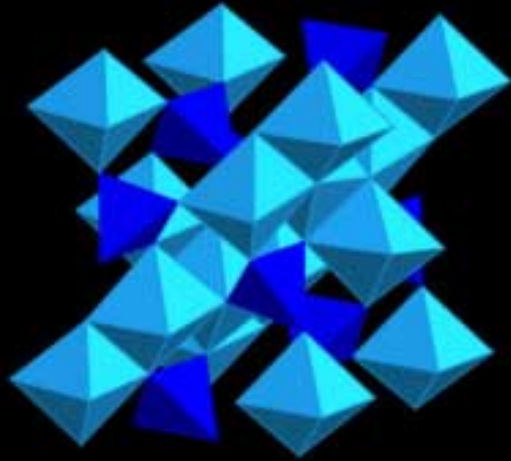




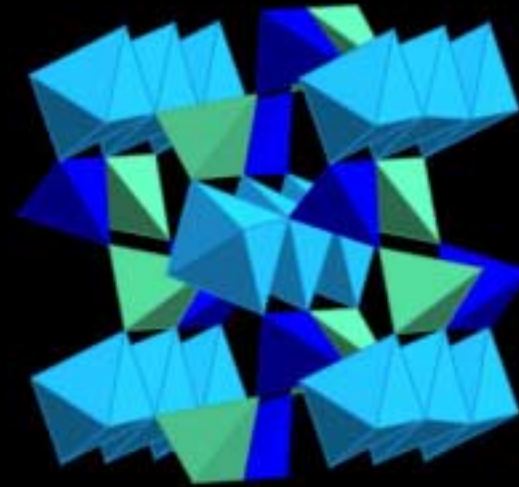
$$\frac{dP}{dT} = \frac{\Delta S}{\Delta V}$$



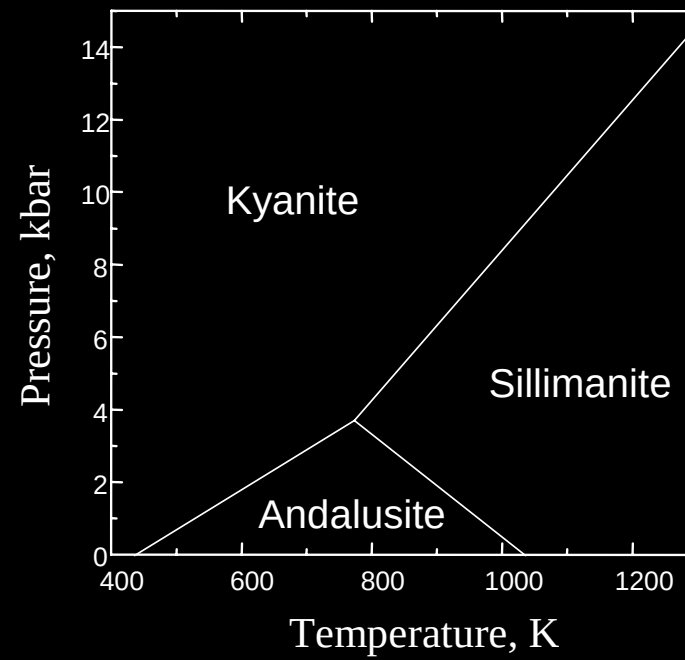
Kyanite



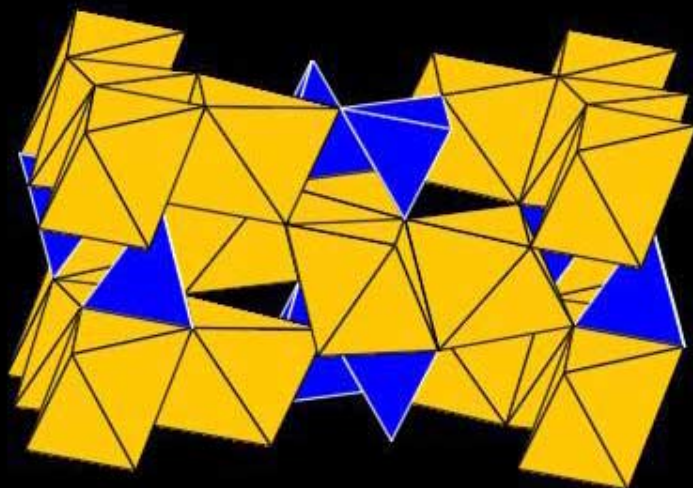
Sillimanite



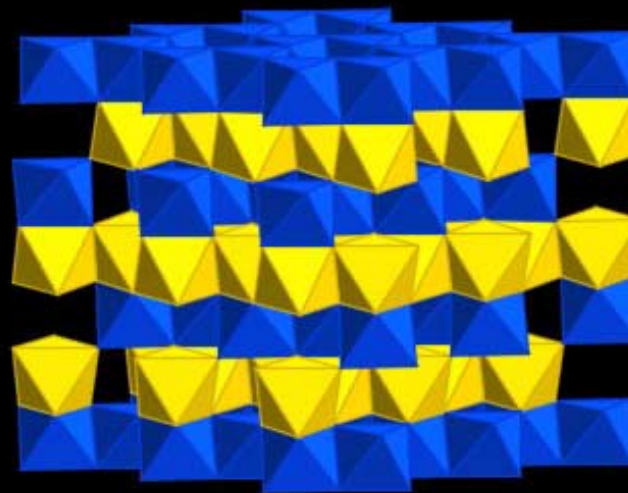
Andalusite



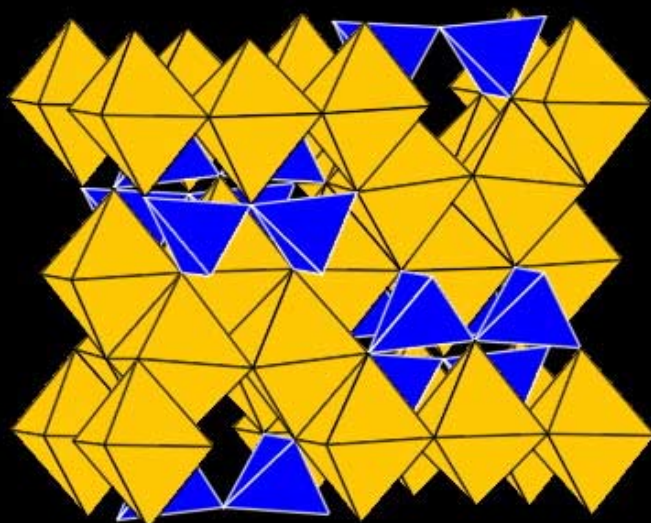
Forsterite



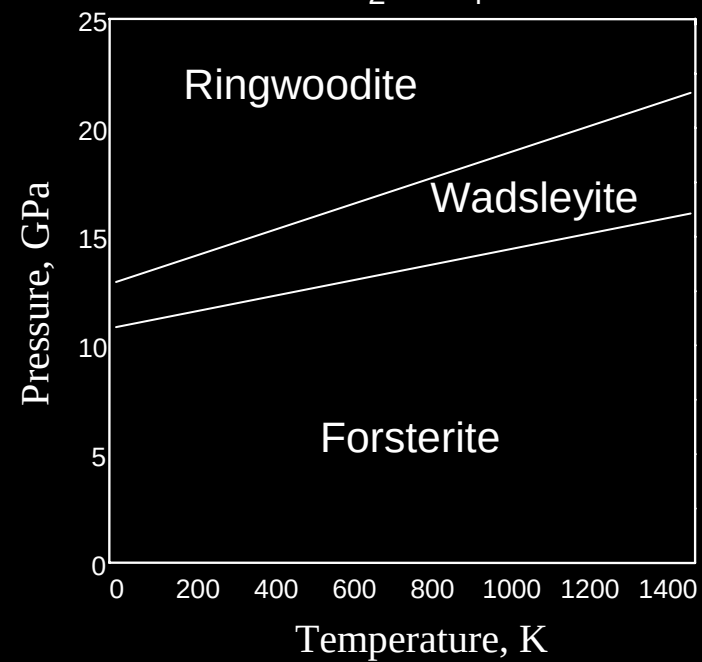
Ringwoodite



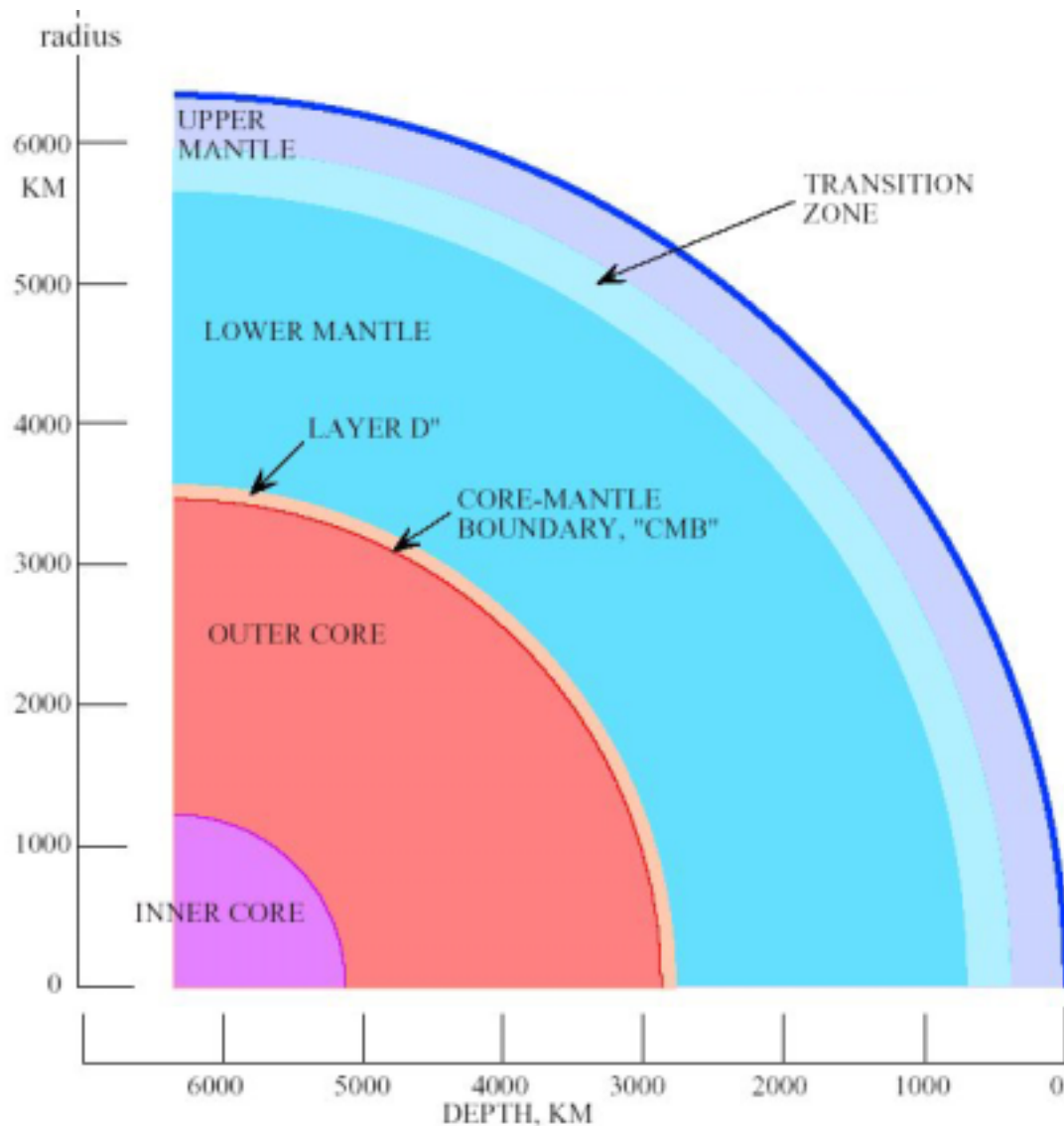
Wadsleyite



Mg_2SiO_4



Seminars (Mondays 14-16)



- (1) Energy minimisation
- (2) C_v , C_p , T -exp, ...
- (3) Fitting potentials
- (4) Calculation of phase diagrams
- (5) Calculation of enthalpies of mixing
- (6) Calculation of disorder
- (7-8) Your own projects
- (9-10) Mol. dynamics

What we need

- GULP manual (available over the net)
- Am. Min. crystal structure data base
- Elastic constants of minerals (Bass, 1995)
- Holland & Powell (1998)
- DL-POLY2 manual
- USB stick

What to read

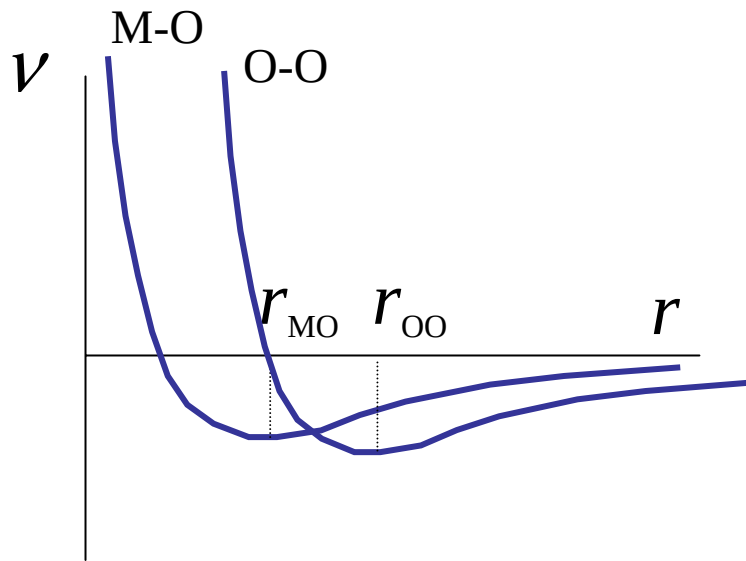
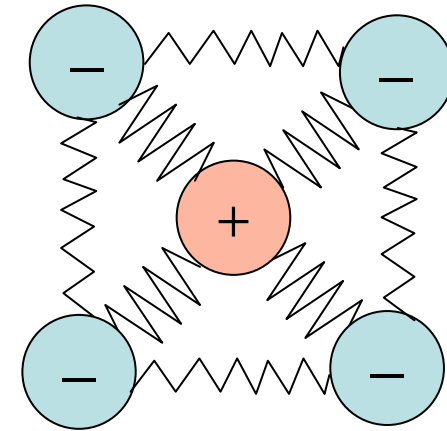
- M.T. Dove. Introduction to Lattice Dynamics. 1993
Cambridge University Press
- J.D. Gale. Simulating the Crystal Structures and
Properties of Ionic Materials from Interatomic Potentials
in Reviews in Mineralogy, vol 42, 2001.

Static Lattice Energy Calculations

Empirical pair potentials

$$v_{ij} = \frac{q_i q_j}{r_{ij}} + \phi_{ij}(r)$$

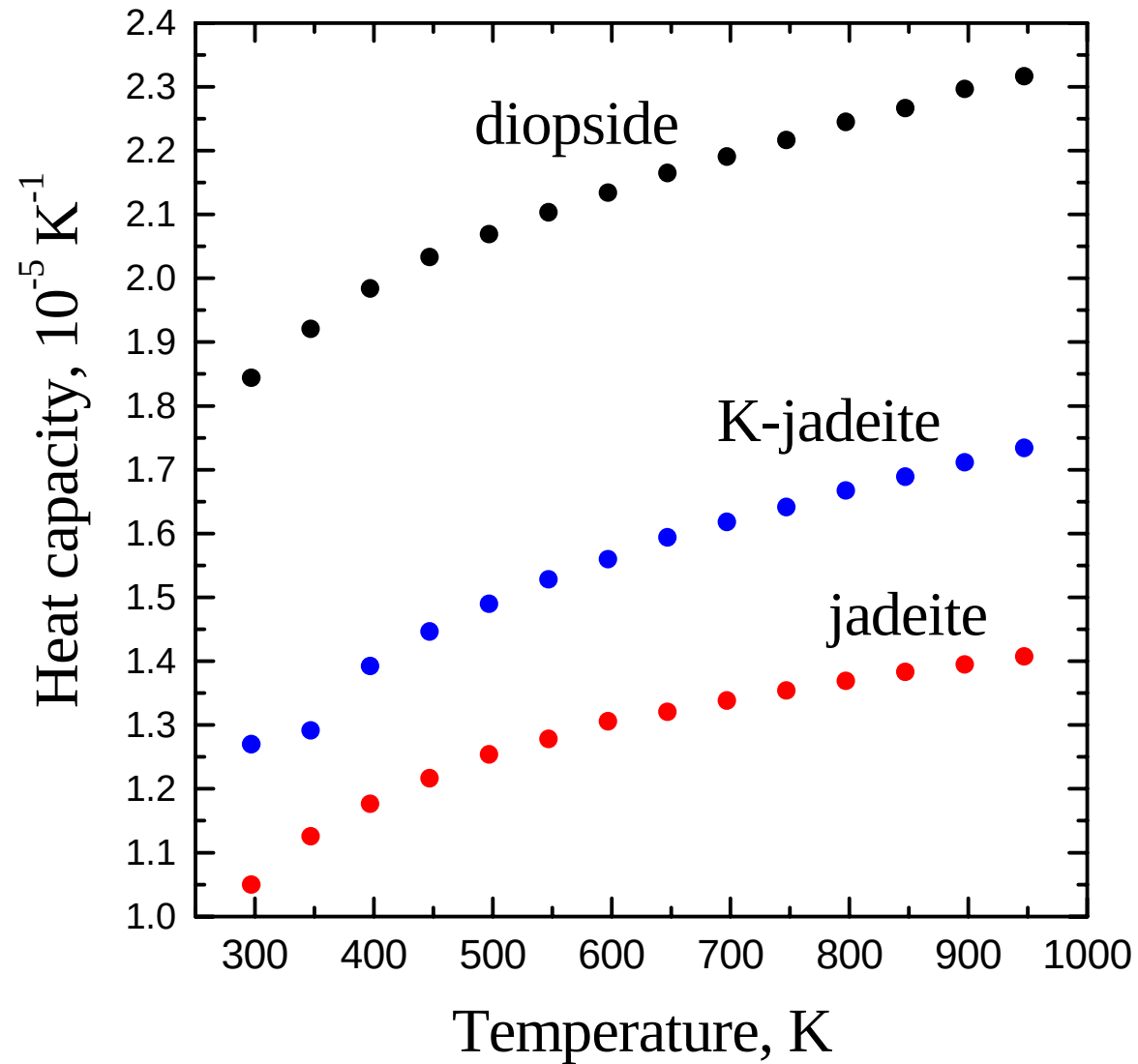
$$E_{static} = 1/2 \sum_{i,j} v_{ij}$$



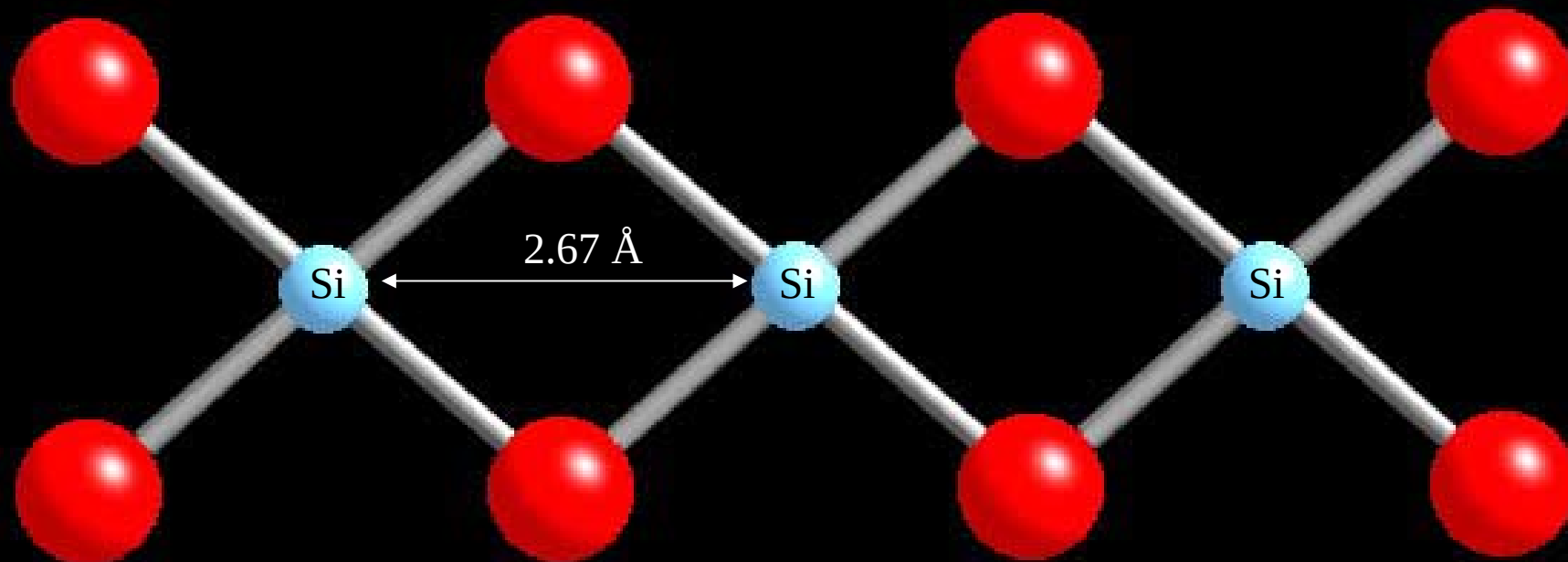
$$\frac{\partial E}{\partial x_i} = 0$$

for all parameters x_i

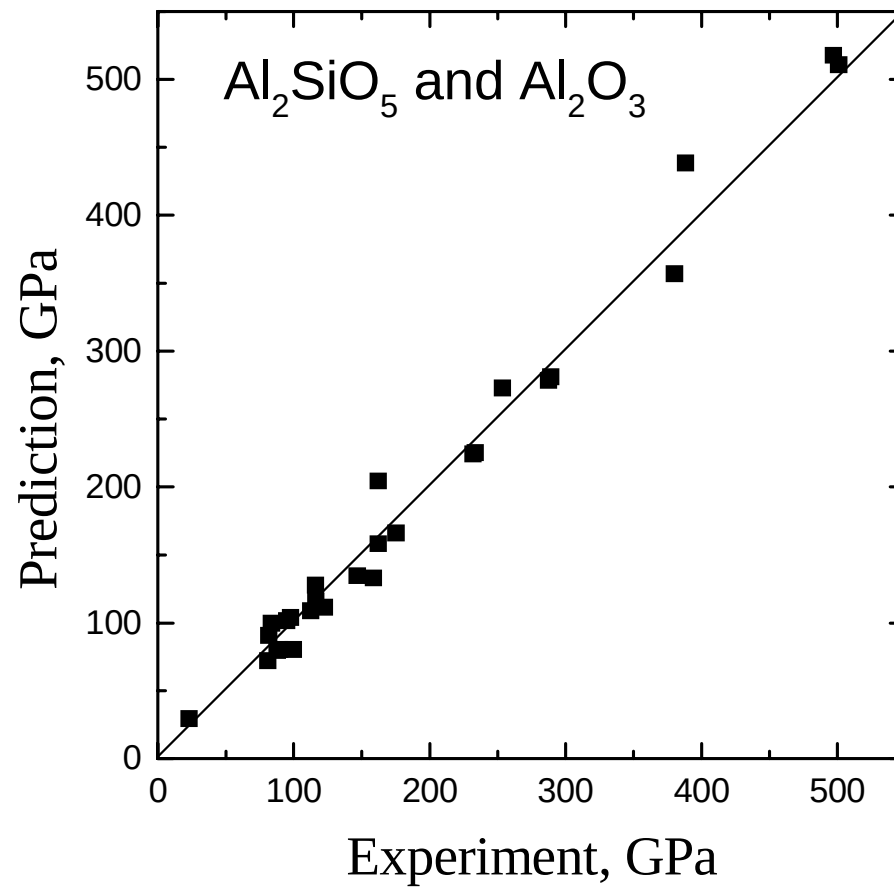
Calculation of thermal expansion



Stishovite

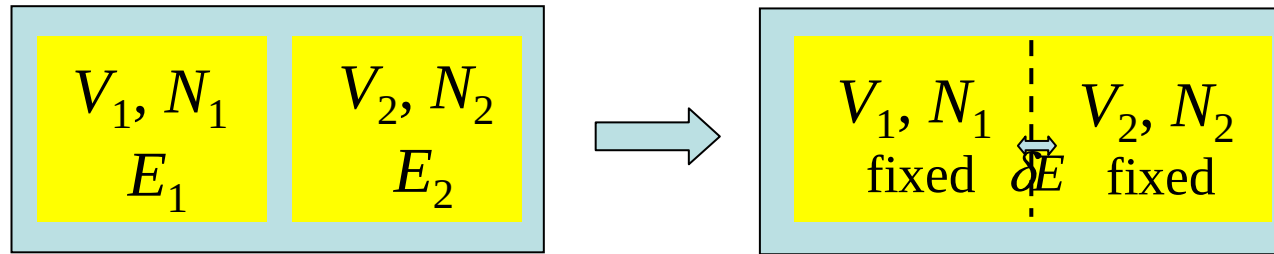


We have good potentials. How can we use them?



Thermal equilibrium

Let us consider two systems brought in thermal contact



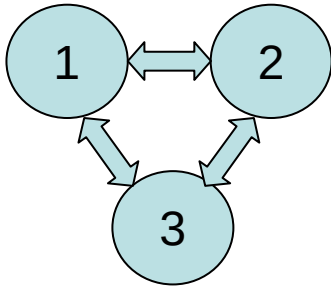
so that the total energy is fixed: $E_0 = E_1 + E_2$

$$dE_0 = dE_1 + dE_2 = 0 \qquad dE_1 = -dE_2$$

The systems will exchange the energy until the equilibrium is achieved. We consider the change in the entropy of the combined system and require:

$$dS_0 = \frac{\partial S_1}{\partial E_1} dE_1 + \frac{\partial S_2}{\partial E_2} dE_2 = 0 \quad \Rightarrow$$

$$\frac{1}{T_1} dE_1 + \frac{1}{T_2} dE_2 = 0 \quad \Rightarrow \quad \left(\frac{1}{T_1} - \frac{1}{T_2} \right) dE_1 = 0 \quad \Rightarrow \quad T_1 = T_2$$



0th law

Thermodynamics

Two systems in thermal equilibrium with the third are in thermal equilibrium with each other

The *temperature* is defined as the quantity which becomes the *same* for thermally equilibrated systems.

1st law

The total energy is conserved

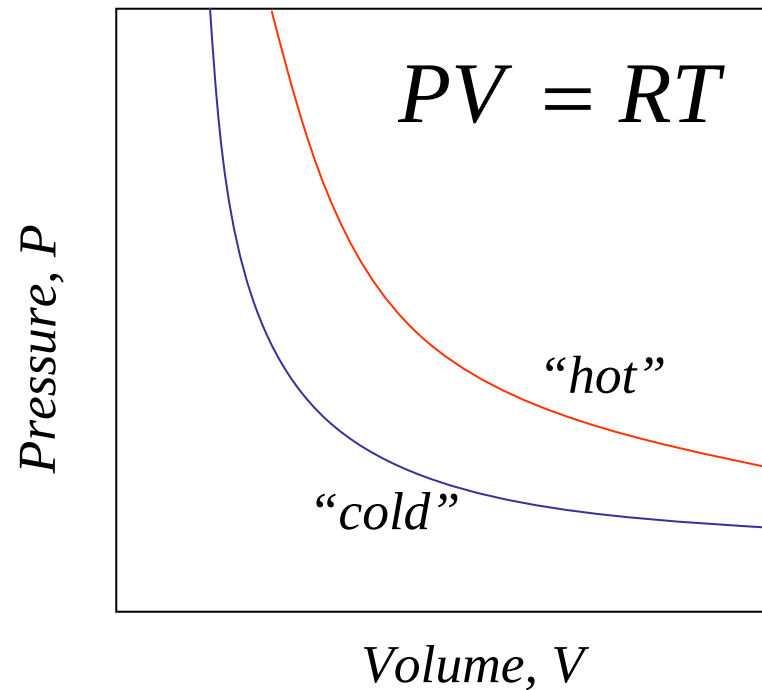
2nd law

Heat flows from a system with a higher T to a system with a lower T



There is a function of state S ,
$$dS = \frac{\delta Q_{rev}}{T}$$

The temperature



$$T = \lim_{P \rightarrow 0} \frac{PV}{(PV)_{\text{triple}}} 273.16 \quad K$$

Only with this choice of the temperature scale the thermodynamics holds true